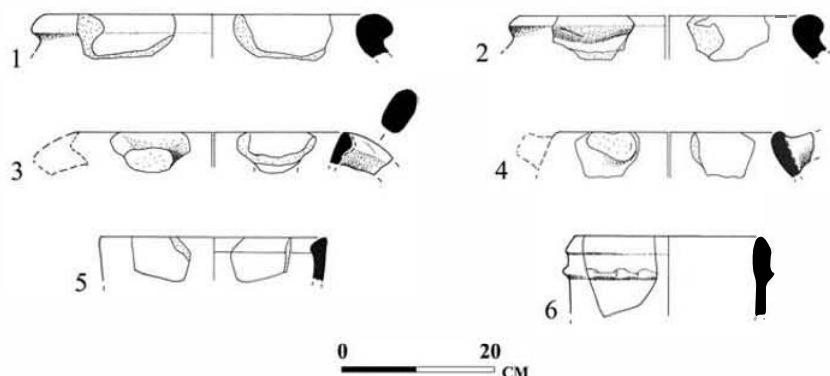




396. Pottery from Wadi Ahmar (8): 1-2. Jars, buff, Chal-EB I; 3-4. Jars, Chal?; 5-6, jars, buff, Mod.

are some unique attributes of this settlement:

- Primarily, it takes special advantage of the boulders and rockfall over the slope. The houses are supported by them with integration of the rocks in the walls. Apparently the settlement location was also chosen for this reason;
- Secondly, the courtyard and broad room (no. 1) resembles the typical Chalcolithic dwelling unit of the nearby Fazeel sites (Bar 2013b; 2014a-b);
- Thirdly, the ratio between enclosures to houses is high, clearly indicating a dwelling site.

Apparently all the above are Chalcolithic, or less probably EB I, but proof is necessary.

Pottery: Chalcolithic/Early Bronze I – 92%; modern – 8%.

Flint: 15 items, see Appendix A.

Previous surveys: none.

Additional Bibliography: Bar 2014a: site 83.

Site 202: 19-16/12/1

MUGHARET SAD KHARIZ

Israel grid: 1910/1624

UTM grid: 7297/5493

Elevation: 120 m b.s.l., 40 m a.s.a.

Name: nearest place

Site type: cave and courtyard

Area: 1 dunam (1000 sq. m)

Topography: slope in a narrow wadi

Rock type: Judea Group

Soil type: stony-desert

Soil quality: 3

Cultivation: none

Cisterns: none

Water source: Fasa'el pool (171),
1 km distantRoad: Fasa'el-Ṭana (C40),
500 m distant

Visit: Dec. 2001; 60 sherds

A large cave on the high rocky slope of Wadi Zamor, close to where it empties into the broad valley of Wadi Ahmar. The cave is about 40 m above the wadi, and the road from the Jordan Valley to Maale Ephraim passes close by to the south of it.

The cave entrance is about 21 m wide, formed by two arches of rock, one in front of the inner one. The cave itself is inside a large rocky strip over a slope, and is about 10 m deep, about 8 m wide, and the ceiling is 2–3 m high. The ceiling is charred, evidence of use by today's shepherds.

In front of the cave is a spacious arch-like courtyard, about 40 m long and 15 m maximum width. The yard is built over a 3–4 m high raised area, forming an even surface, more or less at the level of the opening. In the upper, western, part is an inner partition wall, forming an extra small yard. At the centre is an entrance with standing monoliths at both sides. From the entrance a built pathway descends to the wadi channel.

In the wadi channel below the cave is a built courtyard 7×18 m in size and a cave (not on plan) and modern Bedouin camps.

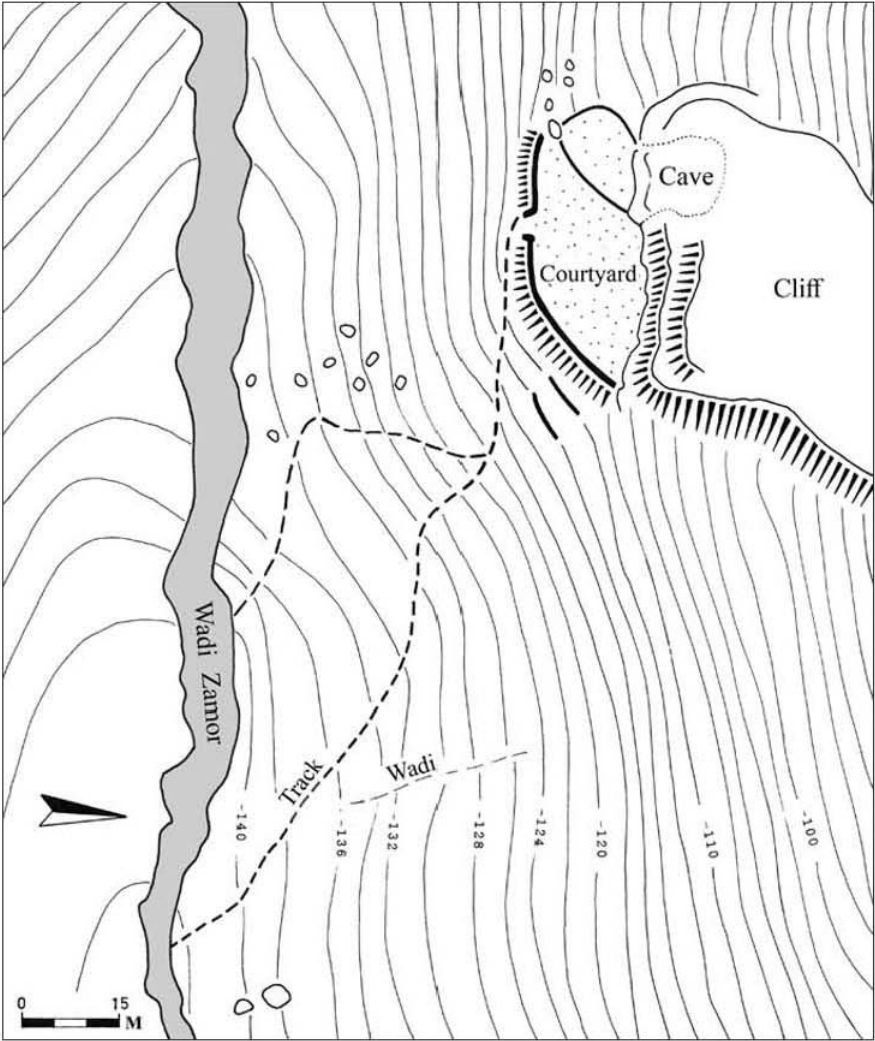
Numerous sherds are scattered in the courtyard and over the slope, and it is apparent that the cave served for dwelling for a long period.

Pottery: Early Bronze I-II – 22%; Late Roman – 24%; Middle Ages – 35%; Ottoman – 19%.

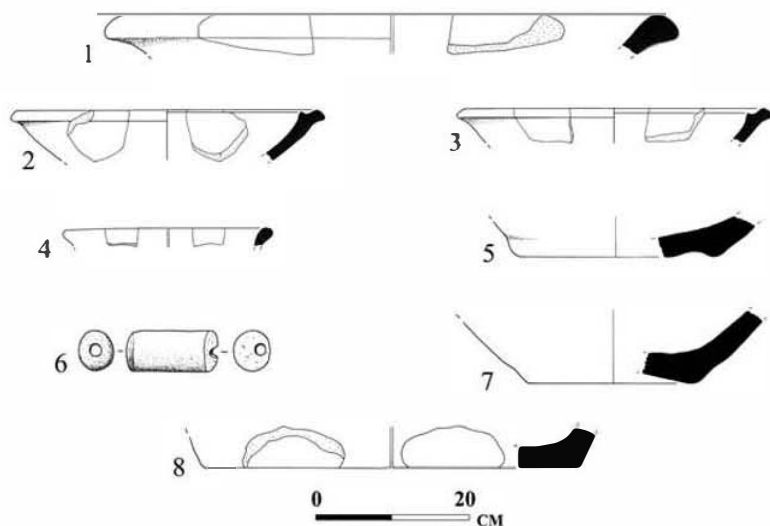
Flint: 44 items, Middle Palaeolithic and Early Bronze, see Appendix A.

Previous survey: none.

Bibliography: Bar 2014a: site 84.



397. Plan of Mugharet Sad Khariz, 2001.



398. Finds from Mughareet Sad Khariz: 1. Bowl, brn, LR; 2, 3. Bowls, buff, MA?; 4. Bowl, buff, EB; 5, 7, 8. Bases, lt brn and buff, EB; 6. Bone bead.



Site 203: 19-16/32/7

'MONUMENT' (2)

Israel grid: 1936/1626

UTM grid: 7323/5495

Elevation: 140 m b.s.l., 100 m a.s.a.

Name: nearest place

Site type: prehistoric

Area: 20 dunams (2 ha)

Topography: slope

Rock type: Judea Group

Soil type: stony-desert

Soil quality: 3

Cultivation: none

Cisterns: none

Water source: Fasa'el pool (171),
3 km distant

Road: Beit She'an–Jericho (B10),
500 m distant

Visits: 1999, repeated visits 2000–2004

A prehistoric site on a slope descending to the Jordan Valley Monument.

Very large scatters of flint items, including workshops, are spread over the entire area, see Appendix A.

Flint: 73 items, Middle Palaeolithic, see Appendix A.

Previous surveys: none.

Site 204: 19-16/32/8

'MONUMENT' (3)

Israel grid: 1937/1622

UTM grid: 7324/5492

Elevation: 150 m b.s.l., 70 m a.s.a.

Name: nearest place

Site type: prehistoric

Area: 20 dunams (2 ha)

Topography: slope

Rock type: Judea Group

Soil type: stony-desert

Soil quality: 3

Cultivation: none

Cisterns: none

Water source: Fasa'el pool (171),
3 km distantRoad: Beit She'an–Jericho (B10),
400 m distant

Visits: 1999, repeated visits 2000–2004

A prehistoric site on the spur of the Jordan Valley Monument.

Very large scatters of flint items, including workshops, are spread over the entire area, see Appendix A.

Flint: 15 items, Upper Palaeolithic, see Appendix A.

Previous surveys: none.

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 ABD = Friedman, D.N., (ed.), 1992. *Anchor Bible Dictionary*, vols. 1-6, New York.
 ADAJ = *Annual of the Department of Antiquities of Jordan*.
 ANET = Pritchard, J.B., (ed.), 1969. *Ancient Near Eastern Texts Related to the Old Testament* (3rd edition), Princeton.
 Ant. = *Josephus, Jewish Antiquities*, with an English Translation by Thackeray, H.St.J., and Marcus, R., (The Loeb Classical Library), London 1966-1969.
 BA = The Biblical Archaeologist.
 BAR = Biblical Archaeology Review.
 BAR Int. Ser. = *British Archaeological Reports International Series*.
 BASOR = *The Bulletin of the American Schools of Oriental Research*.
 BSV = *The Beth Shean Valley, the 17th Archaeological Convention*, Jerusalem (Hebrew).
 CHANE = *Culture and History of the Ancient Near East* (Series).
 CI = Mazar, B., 1980. *Canaan and Israel, Historical Essays*, Jerusalem (Hebrew).
 EB = *Encyclopaedia Biblica*, vols. I-VIII, Jerusalem 1950-1982 (Hebrew).
 EI = *Eretz Israel*.
 ESI = Excavations and Surveys in Israel.
 HA = *Hadashot Arkheologiyot* (Hebrew).
 IAA = Israel Antiquities Authority.
 IEJ = *Israel Exploration Journal*.
 JBL = *Journal of Biblical Literature*.
 JIPS = *Journal of the Israel Prehistoric Society*.
 JMA = *Journal of Mediterranean Archaeology*.
 JPOS = *Journal of the Palestine Oriental Society*.
 JS = Shmueli, A., Grossman, D., and Zeevy, R., (eds.), 1977. *Judah and Samaria, Studies in Settlement Geography*, vols. I-II (*The Land of Israel and the Middle East, Studies in Regional Geography* 1), Jerusalem (Hebrew).
 JSG = Kochavi, M., (ed.), 1972. *Judah, Samaria, and the Golan, Archaeological Survey 1967-1968*, Jerusalem (Hebrew).
 JSP = *Judea and Samaria Publications*.
 JSRS = *Judea and Samaria Research Studies*.
 NEAEHL = Stern, E., (ed.), 1993-2008, *The New Encyclopaedia of Archaeological Excavations in the Holy Land*, vols. 1-5, Jerusalem and Washington.
 PEFQst = *Palestine Exploration Fund Quarterly Statement*.
 PEQ = *Palestine Exploration Quarterly*.
 PJB = *Palaestina Jahrbuch des Deutschen Evangelischen Institut Fuer*

Altertumswissenschaft des Heiligen Landes zu Jerusalem.

PPCNI = Marks, A.E. (ed.), 1976-1983. *Prehistory and Palaeoenvironments in the Central Negev, Israel*, vols. 1-3, Dallas.

RB = *Revue Biblique*.

TA = *Tel Aviv*.

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PART THREE

APPENDICES AND INDICES

APPENDIX A

THE FLINT ASSEMBLAGES

Haim Winter

Prologue¹

The preparation of this Appendix was preceded by quite lengthy deliberations. I hesitated to start before I had answers to two questions concerning ways of solving the problems which I was facing. The answers to these questions would have to be acceptable both to the reader and to me.

The first question is how, if at all, it is possible to retrieve useful information from a few partial finds without any diagnostic artefacts at any single site? Whenever only a few artefacts were present, in most cases they could not provide useful and reliable information for classifying the site. Furthermore, a quantitative, technical report of small flint collections is usually useless, as it does not contribute much to our understanding of the site. For this reason, in this study, in order to utilize the fragmented information of the single site, the alternative was to compare the collection to those of neighbouring sites with similar assemblages. Thus, in some cases it was possible, to some extent, to evaluate the few artefacts found at the single site within the framework of the material culture on a regional basis.

The second question concerns the difficulty of dating collections from early historic sites for the following reasons:

- Some tool types were in use during several successive cultures, thus losing their value as diagnostic markers (*'fossil directeur'*) for a single culture.

¹ This study was carried out more than ten years ago at the level of information available at the time, and published in Hebrew in 2005. Meanwhile research never stopped; several new sites were discovered and we gained new insights into some problems, thus widening the scope and supplying possible answers to open questions. There was no need to re-write the study, as the bulk of the information is still valid. Whenever needed we refer to these new sites and insights in the text or footnotes. Haifa, Spring 2016.

I wish to express my appreciation to Adam Zertal, who provided the artefacts and gave me the opportunity of studying the flint finds in this volume. S. Pinsky added the results on additional sites discovered by the survey team after the Hebrew version was published. A. Ronen helped with information about the Middle and Upper Palaeolithic, B.Z. Begin provided the data about Lake Lisan levels, D. Kaufman helped with information about the Epipalaeolithic, the Manasseh Hill Country Survey team, and ● Cohen helped especially with technical issues, and my son Y. Winter read parts of the study and made important comments. The sketches of the tools were prepared by me and the final drawings were prepared by S. Haad. I am grateful to all of them.

- Statistical methods for the classification of prehistoric assemblages from excavations are not applicable for a few random finds collected from the surface: therefore, in the absence of diagnostic markers, the affiliation of the collection to a definite culture or period is not feasible.

So, whenever diagnostic markers were missing in the flint collection, but sherds which could be dated according to their morphology were present, the alternative was to attribute the dating of the flint collection to that of the pottery assemblage of the site. In this case the collection was usually defined only on the level 'possible' (see Methodology below).

This study is an experiment, maybe innovative and controversial, in contributing to the understanding of the general picture despite the limited and partial data. The possibility of wrong evaluations, especially at multi-period sites, is always present.

Introduction

Years have passed since the publication of the flint assemblages in Volume 2 of the Manasseh Hill Country Survey (Winter 1996; 2008). Even then, we had to deal with the difficulties and limits of analyzing small assemblages collected from the surface. With the experience we have gained, we can now pay more attention to various factors, especially to some of those concerning the area of Volume 4:

- Surface changes caused by natural processes: geological activity, drift and erosion, the rising level of Lake Lisan, and deposition of lake sediments.
- Surface changes caused by human activities: building, agriculture and grazing.
- Objective factors: 1. Vegetation hiding the surface, resulting in selective collection. 2. The difficulty of seeing small items lying on the ground with the naked eye, as opposed to the results of a sieving process common in excavations.

It is quite difficult to draw reasonable conclusions about the way of life of a population from a small assemblage at a single site: however, analyzing several sites from the same period or culture in a specific landscape unit can provide insights into the lifestyle of the palaeo-inhabitants.

Flint artefacts were found in 103 of the 224 sites in this volume. Many of these sites are from early historical times, thus proving the continuity of flint industries utilizing this easily available and cheap material during the Bronze and Iron Ages.

Special attention was paid to the presence of sickle blades in the collections. These tools supply firm evidence of the lifestyle and economy of the population.

Only a few sites which yielded flint artefacts are located at elevations higher than the present day sea level (average sea level as used in the Israel cartographic system), some of these at elevations up to 400 m a.s.l.; but most sites are below the present day sea level. From about 70000 BP to about 12000 BP, the Rift Valley was flooded by the ancient Lake Lisan, which reached its highest level at 164 m b.s.l.. The outstanding phenomenon observed was that below the highest level of the lake (164 m b.s.l.), sites were rare (even though some were as low as 300 m b.s.l.). This phenomenon urged us to examine the influence of the lake level on the possibility of finding prehistoric remains on the present-day surface.

The documentation and analysis of flint artefacts is not an aim in itself, but an additional tool to create an overall picture of the relevant site or culture. As excavations are not feasible at all sites, there is no alternative to a systematic survey to preserve even partial knowledge about the past, but one should refrain from drawing definite final conclusions.

The survey presents the picture according to the various landscape units, and proposes sites for further research and excavation. It exposes whatever is visible on the surface; nonetheless, many secrets are still buried underground waiting to be exposed by conventional excavations. In spite of the limitations of the method, the outcome of this study proves the possibility of drawing interesting conclusions.

Methodology

Our approach to the analysis of the flint assemblages has remained basically unchanged since the flint appendix for Volume 2 was written, except for some differences related to the area, and a new emphasis on some points of the analysis. In contrast to Volume 2, all sites yielding flint artefacts, even in small numbers, are presented here. As mentioned in the Introduction, small assemblages in adjacent sites can contribute to the understanding of the region and the landscape unit. Over the years our awareness that the mere presence of flint artefacts reflects important aspects of the way of life of the population has only strengthened. This is especially true for early historical periods.

For the convenience of the reader the main points of the methodology are presented again here. Whenever possible, we rely on several sources for dating flint assemblages:

1. For typological identifications we used the following sources:

- General type lists: Bar-Yosef 1970; Lee 1973; Hours 1974; Bordes 1961.
- Lists for specific tools, such as arrowheads: Gopher 1985; sickle blades: Burian and Friedman 1979; Gopher 1989a; Rosen 1997.
- Reports from excavations and summarizing studies.

Identification based on typological considerations is possible, but has some disadvantages:

- Different lists were prepared at different times for identical artefacts;
 - Many artefacts do not match any list;
 - Certain tools were used over long periods, and are unsuitable for a chronological evaluation;
 - Regional or local varieties may not match generalized lists;
 - In some cases the typological and radiometric dates do not match (Hovers and Marder 1991: 51–53).
2. Morphological features served as additional tools. Among these were the morphology of cores and industrial waste; the application of the Levallois technique and preparation of the striking platform; the application of the microburin technique; pressure flaking; and the application of the Canaanite blade technology.
 3. At sites where sherds were found, the pottery and flint assemblages were compared in order to determine whether the two assemblages belong to the same period. Whenever diagnostic flint artefacts for other periods were not present the flint could be dated according to the pottery. An example is the Canaanite sickle blade common both in Early and Intermediate Bronze Ages (EBA and IBA). Later studies proved that the Canaanite technology was used as early as the Chalcolithic (see Bar and Winter 2010). A second example is the geometrical sickle blade, in use from the Middle Bronze Age (MBA) to the Iron Ages (IA). The cross-check of pottery and flint data is an important tool, especially for historical periods, when mainly improvised 'ad hoc' flint tools without specific features were produced (Rosen 1997: 34, 158–159).
 4. The compositions of tool assemblages and industrial waste are important characteristics for dating when diagnostic artefacts were not found. In this case the relative shares of the various tool categories in the assemblage and the composition of the industrial waste were analyzed. Two examples are: a relatively high share of scrapers and burins could indicate the Upper Palaeolithic; and Levallois pieces in assemblages without hand-axes could indicate the Middle Palaeolithic (however, recent studies confirm that hand-axes together with Levallois technology existed during the transition between the Lower and Middle Palaeolithic, see Zaidner et al. 2006).

5. Whenever diagnostic markers – characteristic pieces of a certain culture – were present in the assemblage, these served as an important tool for the identification, but we avoided drawing conclusions from a single artefact. Only when the dating was confirmed by other items, mainly by pottery, did we rely on it. The absence or presence of a specific artefact in similar assemblages can possibly define the period.
6. As mentioned above, large areas in the territory of the survey were flooded by Lake Lisan for certain time spans: thus sites which existed prior to the flooding were buried under lake deposits which could accumulate up to 1 m per millennium. Hence it is impossible to find sites earlier than the Epipalaeolithic period on the present-day surface, unless they were submerged only for short periods, or exposed either by natural processes or by human activities.
7. Furthermore, the flint artefacts under discussion did not originate from an organized excavation; they were collected from the surface; hence we refrained in most cases from drawing definitive conclusions about the single site.

For this purpose the identification of the periods was divided into three categories:

- Highly reasonable identification
- Reasonable identification
- Possible identification

Some of the collections did not provide any indicative data and these sites were categorized as “not defined”.

For the analysis of the finds we relied mainly on data from sites identified as highly reasonable or reasonable.

The following tools and technologies served for the identification process of cultures along the timeline:

Lower Palaeolithic period:

- Bifacial hand-axes identified the period. Tools which are similar to some extent, were found in Neolithic sites, but these usually showed battered lateral edges (Olami 1984: 139, 151-152; Ronen 1984: 163 and figs. 10.29.1, 10.30.1). No Lower Palaeolithic sites were found in the studied area.

Middle Palaeolithic period:

- Presence of the Levallois technique at open air sites without hand-axes in the collection. In this country hand-axes are not generally present in Middle Palaeolithic assemblages; this is different from Europe (Goren-Inbar and Belfer-Cohen 1998: 206);

- Preparation of the striking platform and the presence of '*chapeau de gendarme*' bases;
- Relative abundance of side-scrapers;
- Relative percentage of the various tool groups (Bordes 1961).

Upper Palaeolithic period:

- Relative shares of end-scrapers and burins, and especially the presence of carinated scrapers;
- Relative abundance of blades and the presence of a microlithic component.

Epipalaeolithic period:

- Morphology of cores and waste materials;
- Relative share and the morphology of the various types of microliths. The phases within the period can be defined by these traits;
- Morphology of end-scrapers and burins;
- Presence of sickle blades.

Neolithic period:

- Presence of a microlithic component;
- Morphology of bifacial tools (axe, adze and chisel);
- Presence of naviform cores;
- Presence of various types of arrowheads;
- Presence of various types of sickle blades and reaping knives.

Wadi Rabah phase:

- Morphology of sickle blades;
- Morphology of bifacial tools.

Chalcolithic period:

- Morphology of bifacial tools;
- Morphology of sickle blades;
- Presence of perforated flint discs;
- Presence of fan-scrapers.

Early Bronze Age (EBA) and Intermediate Bronze Age (IBA):

- Presence of Canaanean blades and Canaanean sickle blades;
- Presence of fan-scrapers.

Middle Bronze Age (MBA), Late Bronze Age (LBA) and Iron Ages (IA I; IA II):

- Presence of large geometric sickle blades.

In sites with few artefacts special attention was paid to striking platforms, bases and bulbs of percussion. These indicate intentional production of the artifact, even when not all the components of the industry were found.

Definitions:

This study describes the finds in general terms; hence we have limited ourselves to dividing the collections into two categories:

1. Artefacts without retouch: all the products and waste materials without any further treatment produced by the initial flaking process. Flakes and blades which could serve for various tasks even without retouch were included in this category.
2. Tools and retouched pieces: all the artefacts with secondary treatment (such as retouch) shaped into tools for various specific tasks.

Due to the small assemblages collected at random from the surface and the difficulty of distinguishing between the stages of a particular culture, in many cases the period could be defined only in general terms, e.g. occasionally the stages of the Neolithic or the Iron Ages are not specified.

Description of sites with flint assemblages:

Landscape Unit 27

Site 1. Wadi Qa'un (1)

Israel Grid 1931/2011, Elevation 25 m b.s.l.

Artefacts without retouch and waste: Blade 1.

A single, intentionally detached blade cannot be affiliated with any culture or period, but the pottery sherd collection was from the IBA.

Period: IBA – possible.

Site 2. Khirbet Qa'un (1)

Israel Grid 1941/2016, Elevation 45 m b.s.l., 177 flint artefacts (Fig. A1).

Waste and artefacts without retouch: Bladelet core 3; Chunk 56; Chip 51; Flake 28; Blade 11; Bladelet 3. Total 152.

Tools and retouched artefacts: Side-scraper 1; Burin 5; Awl 3; Retouched flake 4; Retouched blade 2; Retouched bladelet 1; Retouched microlith 2; Notch 5; Varia 2. Total 25.

The pottery collection was dated to the EBA I, but no indications of this period were found in the flint assemblage. However, several artefacts indicate a microlithic industry e.g. small waste materials, bladelet cores, bladelets and two retouched microliths (one broken), but diagnostic artefacts for the stages of the period were missing. However, at least a part of the flint assemblage could originate from the EBA.

Periods: Epipalaeolithic – reasonable identification (part of the collection); EBA – possible identification (part of the collection).

Site 4. Cemetery of Qa'un

Israel Grid 1942/2013, Elevation 50 m b.s.l.; 45 flint artefacts.

Waste and artefacts without retouch: Primary flake 1; Chunk 7; Chip 11; Flake 15; Blade 4. Total 38.

Tools and retouched artefacts: Burin 4; Sickle blade (backed) 1; Notch 2. Total 7.

The pottery collection, originating from various periods (Chalcolithic to IA), makes it difficult to identify the flint collection. Backed sickle blades are common in the Chalcolithic, and it makes sense that at least a part of the flint assemblage originates from this period.

Period: Chalcolithic – reasonable identification (part of the collection).

Site 5. Khirbet Qa'un (2)

Israel Grid 1942/2014, Elevation 45 m b.s.l.; 4 flint artefacts (Fig. A1).

Waste and artefacts without retouch: Primary flake (blade) 1; Flake 1.

Tools and retouched artefacts: Awl 1; Axe 1.

Sherds: 84% were from the IBA, and 10% from the Chalcolithic. Generally there were no axes in the IBA, but the other items are common in both periods.

Periods: Chalcolithic – highly reasonable identification (part of the collection); IBA – reasonable identification (part of the collection).

Site 12. Tell Tzaf

Israel Grid 2019/2014, Elevation 280 m b.s.l.

First publication – Gopher 1989b. This publication analyzes the flint assem-

blages from a rescue excavation. According to the pottery and flint finds the site was dated to the middle of the 5th millennium BC – Wadi Rabah and Middle Chalcolithic stages.

Periods: Wadi Rabah – highly reasonable identification (in this study Wadi Rabah stage was considered Neolithic); Chalcolithic – highly reasonable identification.

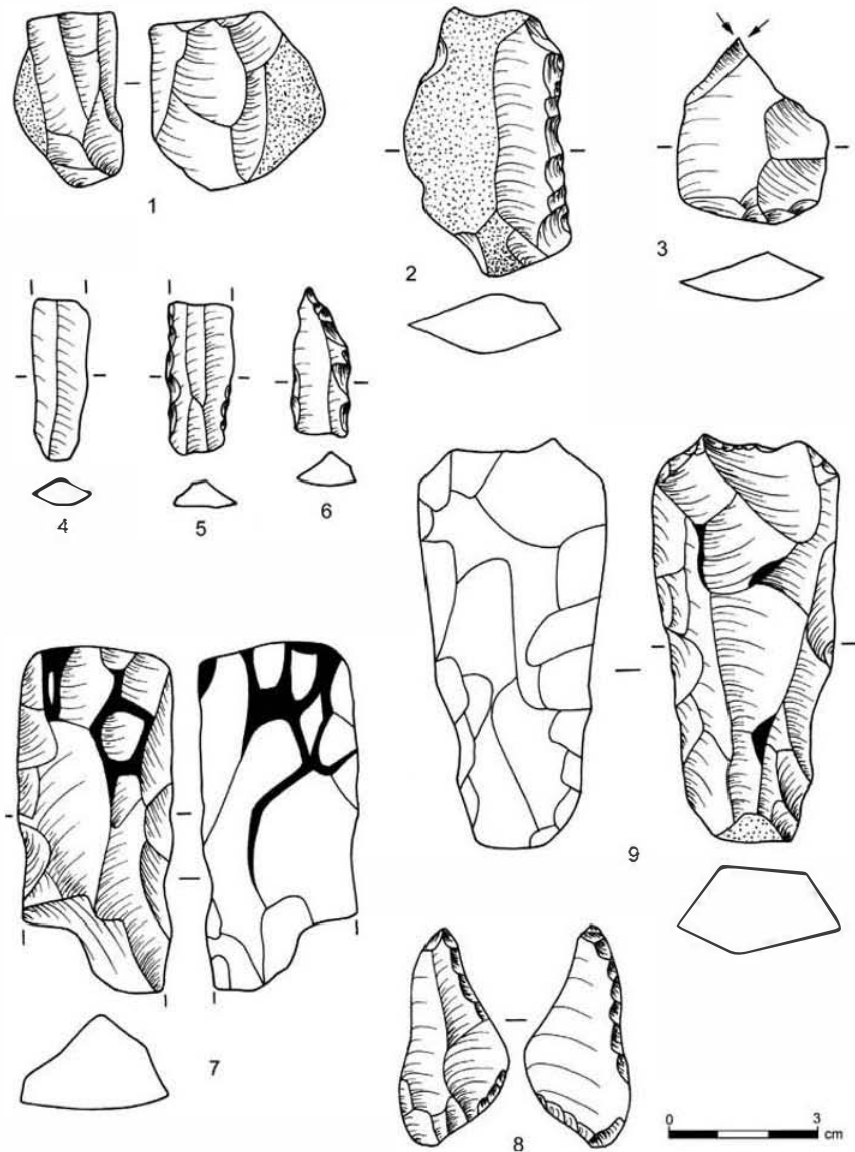


Fig. A1. Kh. Qa'un (1): 1-Core; 2-Side-scraper; 3-Burin; 4-Bladelet; 5-Retouched blade; 6-Microlith with arched back. 'Ein Buleibil' 7-Adze; 8-Borer. Kh. Qa'un (2): 9-Axe.

Site 23. 'Ein ed-Deir

Israel Grid 1993/1995, Elevation 196 m b.s.l.; 6 flint artefacts.

Waste and artefacts without retouch: Core 1; Flake 1.

Tools and retouched artefacts: Awl 1; Axe (fragment) 1; Adze (fragments) 2.

The pottery from this site originates from the MBA to the IA. The axe and the adze are missing from assemblages of these periods, but are present in the Neolithic and Chalcolithic. The other items can be attributed to all the periods mentioned above. The adzes and axes with trapezoidal cross-sections are characteristic of the Chalcolithic (see details in the concluding Discussion). The Chalcolithic period is definitely present at 'Ein Buleibil (Site 24), about 700 m from this site, and at Tell ed Deir (2)(Site 27). Perhaps some artefacts from these sites reached this site; or alternatively there was human activity, not documented by sherds, at the site during the Chalcolithic.

Periods: Chalcolithic – highly reasonable identification (part of the collection).

Bronze and Iron Ages – possible identification (part of the collection).

Site 24. 'Ein Buleibil

Israel Grid 1999/1996, Elevation 220 m b.s.l.; 37 flint artefacts (Fig. A1).

Artefacts without retouch: Core 2; Primary flake 5; Core trimming element 1; Chunk 3; Flake 9; Blade 3. Total 23.

Tools and retouched artefacts: Transversal scraper 1; Awl 1; Borer 1; Retouched flake 1; Notch 4; Denticulate 2; Polished adze 2; Varia 2. Total 14.

The flint finds match the sherd collection: 95% of these are from the Chalcolithic. Similarly to other sites of the period (e.g. Site 27 below), most flint artefacts are made from light-coloured medium-grained flint. The assemblage includes cores, primary elements, core trimming elements, chunks, flakes, blades and retouched tools. All these present the whole sequence of an organized local flint industry in the Chalcolithic. This dating is further confirmed by the polished adzes with trapezoidal cross-sections (see details in the concluding Discussion).

Period: Chalcolithic – highly reasonable identification.

Site 26. Tell el-Beidha-Shamsiyeh

Israel Grid 1978/1986, Elevation 170 m b.s.l.

Three flint artefacts without retouch: Flake 2; Blade 1.

The pottery assemblage, originating from different surveys, is the earliest

and largest collection attributed to the EBA; hence the flint finds are similarly attributed to this period.

Period: EBA – reasonable identification.

Site 27. Tell ed-Deir (2)

Israel Grid 1998/1991, Elevation 215 m b.s.l.; 12 flint artefacts (Fig. A2).

Artefacts without retouch: Core 1.

Tools and retouched artefacts: Burin 1; Retouched flake 1; Retouched blade 1; Sickle blade 2; Notch 2; Adze 3; Varia 1. Total 11.

Similarly to other contemporary sites (e.g. Site 24), most artefacts are made from light-coloured medium-grained flint. The pottery finds are from late historical periods, but the adzes with trapezoidal cross-sections and the sickle blades confirm a Chalcolithic date.

Period: Chalcolithic – highly reasonable identification.

Site 28. Tell ed-Deir (1)

Israel Grid 1999/1986, Elevation 200 m b.s.l.

Four retouched artefacts: Varia 4.

Only a few of the sherds are from the Chalcolithic and EBA, and the rest are from later historical periods. The few flint artefacts were broken, and their original shape could not be identified. All were retouched and had use wear signs. Apparently they should be attributed to the Chalcolithic or EBA.

Period: Chalcolithic/EBA – possible identification.

Site 31. ej-Jobeh

Israel Grid 2006/1977, Elevation 200 m b.s.l.; 39 flint artefacts.

Pottery sherds, basalt tools and flint tools found at this site were all from the Chalcolithic; among them were three typical Chalcolithic adzes. When this Appendix was prepared the artefacts could not be located.

Period: Chalcolithic – highly reasonable identification.

Site 33. Tell Abu Sus

Israel Grid 2029/1978, Elevation 250 m b.s.l.; 28 flint artefacts (Fig. A2).

Artefacts without retouch: Core 1; Primary flake 4; Chunk 1; Flake 11; Blade 1; Canaanan blade 1; Bladelet 1. Total 20.

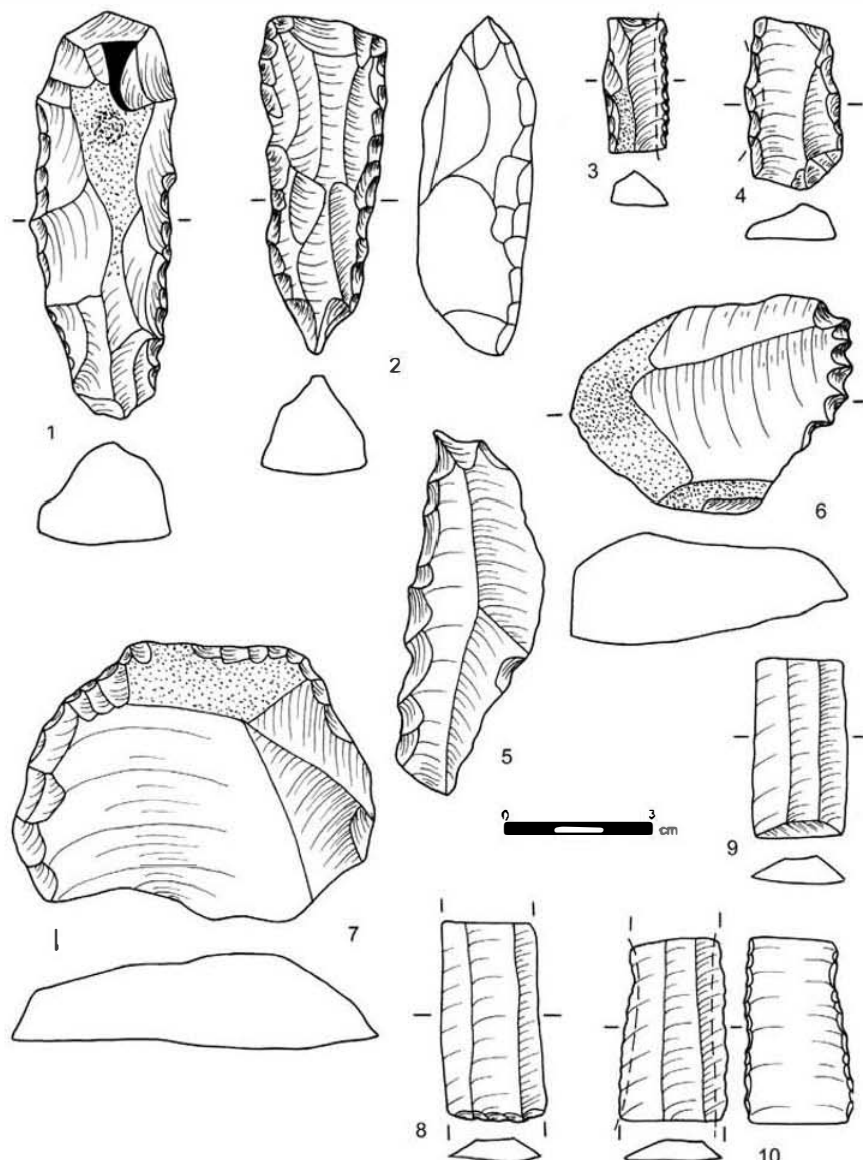


Fig. A2. Telled-Deir (2): 1-Adze; 2-Axe; 3, 4-Sickle blades. Tell Abu Sus: 5-Awl; 6-Denticulate; 7-Rounded scraper; 8, 9-Canaan blades; 10-Canaan sickle blade.

Tools and retouched artefacts: Rounded scraper 1; Awl 1; Retouched blade 2; Canaanean sickle blade 1; Denticulate 1; Truncation 1; Varia 1. Total 8.

The pottery finds are mostly from proto-historical and early historical periods, from the Chalcolithic to IA II. Most flint artefacts are made from light brown to brown good quality flint. The Canaan blade and the Canaan sickle blade should be attributed to the EBA. The rounded scraper of beige calcareous flint could originate from the Chalcolithic, and some other artefacts from still other periods.

Period: EBA – highly reasonable identification (part of the collection).

Summary of Landscape unit 27

356 flint artefacts were collected at 11 sites in this landscape unit (the data for the 12th site - Tell Tzaf is from Gopher 1989b).

Seven sites were attributed to the Chalcolithic at levels of highly reasonable and reasonable identification and one site at the level of possible. Apparently, at this period, a cluster of settlements populated the area (see Bar 2014 for further details). The climate and soil conditions allowed extensive agriculture

Landscape Unit 27. Summary of sites by period. Identification according to flint artefacts			
Period	Highly reasonable	Reasonable	Possible
Middle Palaeolithic	-	-	-
Upper Palaeolithic	-	-	-
Epipalaeolithic	-	1	-
Neolithic	1	-	-
Chalcolithic	6	1	1
Early Bronze (EBA)	1	1	2
Intermediate Bronze (IBA)	-	1	1
Middle Bronze (MBA)	-	-	-
Late Bronze (LBA)	-	-	-
Bronze general (BA)	-	-	1
Iron I (IA I)	-	-	-
Iron II (IA II)	-	-	-
Iron general (IA)	-	-	1
Not defined	-		

Table A1: Landscape unit 27 - Periods and identification level.

and cultivation of grain, an important component of the economy. The presence of sickle blades and celt tools confirms this assumption.

From the EBA one site was defined as highly reasonable, one reasonable and two possible. At Site 2, Khirbet Qa'un (1), a pottery collection and quite a large flint collection from this period were found. Because of the presence of bladelet cores and a few microliths it seemed reasonable to attribute the flint to the Epipalaeolithic. The composition of the assemblage did not allow such a definite conclusion, and at least a part of the collection should be attributed to the EBA.

From the Epipalaeolithic one site was identified as reasonable and from the Neolithic (Wadi Rabah) one as highly reasonable. Two sites were attributed to the IBA, reasonable and possible identifications. The few sites from each period, the small assemblages and the character of the collections do not allow evaluations and conclusions about activities and economy at these sites.

During all these periods the level of Lake Lisan was lower than the elevation of the surveyed sites; thus it was possible to discover the finds on the surface (Bar Tov et al. 2002: fig. 7). No flint artefacts from the Middle or Upper Palaeolithic periods were found, because these, if present, would have been buried under the lake deposits (Bar Tov et al. 2002).

Landscape Unit 28

Site 37. Habs Qatwi (2)

Israel Grid 2003/1915, Elevation 160 m b.s.l.

Two retouched artefacts: End-scraper 1; Awl 1.

Most sherds are from late periods. Only 10% are from IA II, and tentatively the flint originates from this period. The importance of reporting such a small assemblage is its statistical value as a part of other contemporary sites in the area.

Period: IA II – possible identification

Site 41. Kh. Wadi Sadd el-Balqawi

Israel Grid 2030/1840, Elevation 290 m b.s.l.; 4 flint artefacts (Fig. A6).

Artefacts without retouch: Blade 1.

Tools and retouched artefacts: Rounded scraper 1; Side-scraper 1; Canaanan sickle blade 1. Total 3.

Only a few artefacts were found. All sherds were attributed to the IBA, hence the flint assemblage should be attributed similarly to this period. The Canaanan sickle blade confirms this assumption.

Period: IBA – highly reasonable identification.

Site 43. El Mudaraj (1)

Israel Grid 2009/1792, Elevation 100 m b.s.l.; 4 flint artefacts.

Artefacts without retouch: Flake 2.

Tools and retouched artefacts: Retouched flake 2.

80% of the pottery were from IA I, 10% were from IA II, and the rest was from late periods. Hence it makes sense to attribute the artefacts mainly to IA I. Periods: IA I – reasonable identification.

IA II – possible identification.

Site 47. Huzuq Musa

Israel Grid 2014/1782, Elevation 260 m b.s.l.; 2227 flint artefacts (Figs. A3, A4, A5; Tabs. A2, A3).

The site was discovered in 1989, and at first only a few flint artefacts, including several bladelet cores and some other non-diagnostic items, were collected; so the identification of the assemblage was impossible. Numerous deep conical mortars, similar to those from Natufian sites at Nahal Oren (Stekelis and Yizraeli 1963: 11–12) and Rosh Zin (Henri 1976), were found. These enigmatic installations called for further research. In Autumn 2004 it was decided to survey the site thoroughly in order to collect more artefacts for identification. A dense concentration of flint artefacts was found on an eastward-sloping terrace. On top of the terrace were two large boulders forming a kind of gateway, and between them some smaller boulders, perhaps the roof of a collapsed cave. Downhill, to the north, east, and south, the density of flint decreased, and flint finally disappeared. On the eastern slope structures similar to those of Site 175 were found, and on the rocky surface conical mortars were dominant. The thorough survey yielded over 2000 artefacts, a very special assemblage. Most items were of good quality brown-violet flint originating from pebbles without cortex. A few artefacts, including one fan scraper from the Chalcolithic or EBA, and probably one side-scraper, had a thick calcareous cortex. Apparently these were not a part of the main assemblage.

The large sample enabled an analysis according to the procedure used at excavated sites, and the data are presented in Tables A2 and A3. We used the

typological list for the Epipalaeolithic of the Levant (Hours 1974) in order to find similarities to the special composition of our assemblage.

The sorting yielded numerous bladelet cores and retouched microliths, but these were in relatively low numbers compared to the number of cores (the number of geometric microliths was negligible). In a surface survey such small items could be overlooked, but in our case many other small artefacts such as chips and micro-borers were collected. Thus, apparently, the number of microliths approximately represents their real share in the assemblage.

Special attention had to be paid to the lack of bifacial tools and projectile points of any kind (one was found) and the relatively high numbers of borers, backed blades, and sickle blades. The latter appear first in the Natufian stage, but the absence of geometric microliths and lunates specifically prevented us from defining the site as Natufian. Arrowheads appeared first in the Pre-Pottery

	Huzuq Musa		Nahal 'Ein Gev II
	N	%	%
Waste			
Chips	336	62.2	-
Chunks	204	37.8	-
Waste Total	540	100	-
Artefacts without retouch			
Flakes	535	43.25	54.6
Blades	250	20.21	13.8
Bladelets	145	11.72	10.9
Micropoint	15	1.21	-
Primary elements	167	13.50	10.1
CTE	7	0.57	6.8
Carinated blades	40	3.23	0.4
Cores	49	3.96	2.5
Burin spalls	29	2.34	0.9
Artefacts without retouch Total	1237	100	-
Special artefacts	N		
Microburins	11*	-	8
Hammerstones	4	-	-
Artefacts less tools Total	1792		

*All artefacts have a typical notch, but the definition as microburin is uncertain.

Table A2: Huzuq Musa, waste and artefacts without retouch.

Neolithic A, and their absence here did not allow defining the assemblage as originating from this stage. Apparently the site had to be dated to the time between these two stages. It should be remembered that the combination of a high number of awl-borers together with sickle blades was present in many Neolithic sites in the area (Netiv Hagdud, Gilgal 1, Gilgal 3, Gilgal 4 – summary of the data in Groman-Yaroslavski 2003: tables 7, 25, 52, 95); but these were always found together with arrowheads, bifacial tools, and microliths. An overview of publications on late Epipalaeolithic sites made it clear that the assemblage of Huzuq Musa is comparable to only one site close to Lake Galilee (Kinneret) – Nahal 'Ein Gev II (Bar-Yosef and Belfer-Cohen 2000). The similarity of the sites is noticeable due to the composition of the assemblages – the dominance of borers, backed blades, sickle blades, and also technological aspects. At both sites abrupt retouch was applied for backing of blades and sickle blades (some of these had curved backs), and the cutting edges were not retouched (except for two serrated sickle blades). Both lateral sides of the borers were shaped by

	Huzuq Musa		Nahal 'Ein Gev II
	N	%	%
A - Side-scrapers	6	1.38	
B - Scrapers total	29	6.67	3.1
D - Burins	34	7.82	4.2
E - Borers and Awls	102	23.45	23.3
F - Backed Blades	25	5.75	3.5
F 6 - Sickle Blades	45	10.34	4.9
G - Truncations	18	4.14	4.2
H 1, 2 - Notches	69	15.86	10.8
H 3 - Denticulates	11	2.35	
I - Point on Blade	2	0.46	0.4
J - Retouched Artifact	46	10.57	8.3
K - Multipurpose Tool	7	1.61	0.7
L - Varia	12	2.76	14.2
M - Non-Geometric Microliths	25	5.75	16.3
N 3, 4 - Trapezes, Rectangles - Geometric Microliths	1	0.23	1.7
N 6 - Geometric Microliths - Lunates	3	0.69	4.5
Total Tools	435	100	100

Table A3: Huzuq Musa, the tools (type list after Hours 1974; Nahal Ein Gev 2 data after Bar-Yosef and Belfer-Cohen 2000).

abrupt retouch, and the majority was within the dimensional limits of microliths. The absence of arrowheads and bifacial tools and the small numbers of microliths completed the picture.

Nahal 'Ein Gev II was dated to the transition between the Natufian and the Pre-Pottery Neolithic A (Bar-Yosef and Belfer-Cohen 2000: 67–68). The authors even considered the notion that the site belonged to a wider cultural entity located on both sides of the River Jordan (Bar-Yosef and Belfer-Cohen 2000: 67). Apparently our site confirms this assumption. It seems reasonable to give this new entity a name: 'Lisanian Culture' seems appropriate. Chronologically,

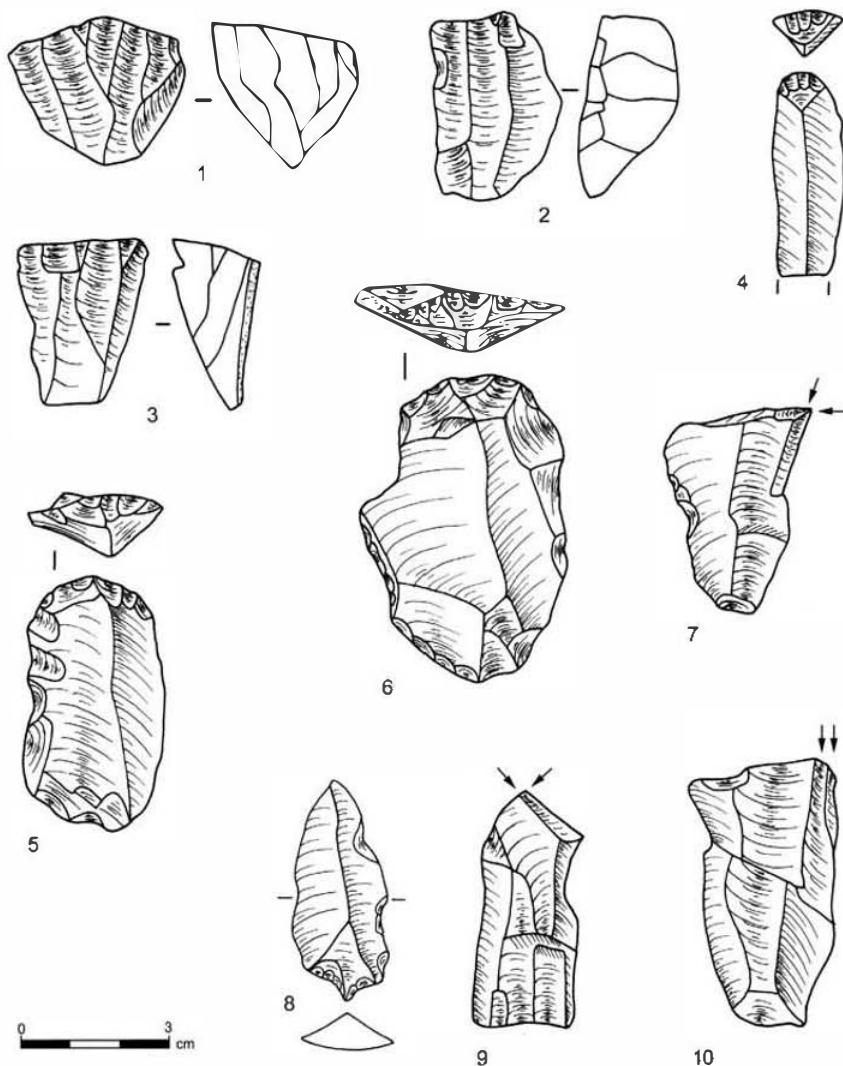


Fig. A3. Huzuq Musa: 1-3-Cores; 4-6-End-scrapers; 7, 9, 10-Burins; 8-Arrowhead.

this culture is contemporary with the regional 'Harifian Culture' in the Negev. According to local conditions the economy of this culture was mostly based on hunting and gathering, as indicated by the absence of sickle blades and the presence of 'Harifian Points', which apparently served as projectile points. The economy of Huzuq Musa and Nahal 'Ein Gev II was similar. Both sites are located on terraces above streams flowing into Lake Lisan. The humid environment apparently supported an economy based on gathering of wild cereals. The abundance of sickle blades at both sites seems to prove this assumption. It should be noted that wear-lustre also accumulates on sickle blades when

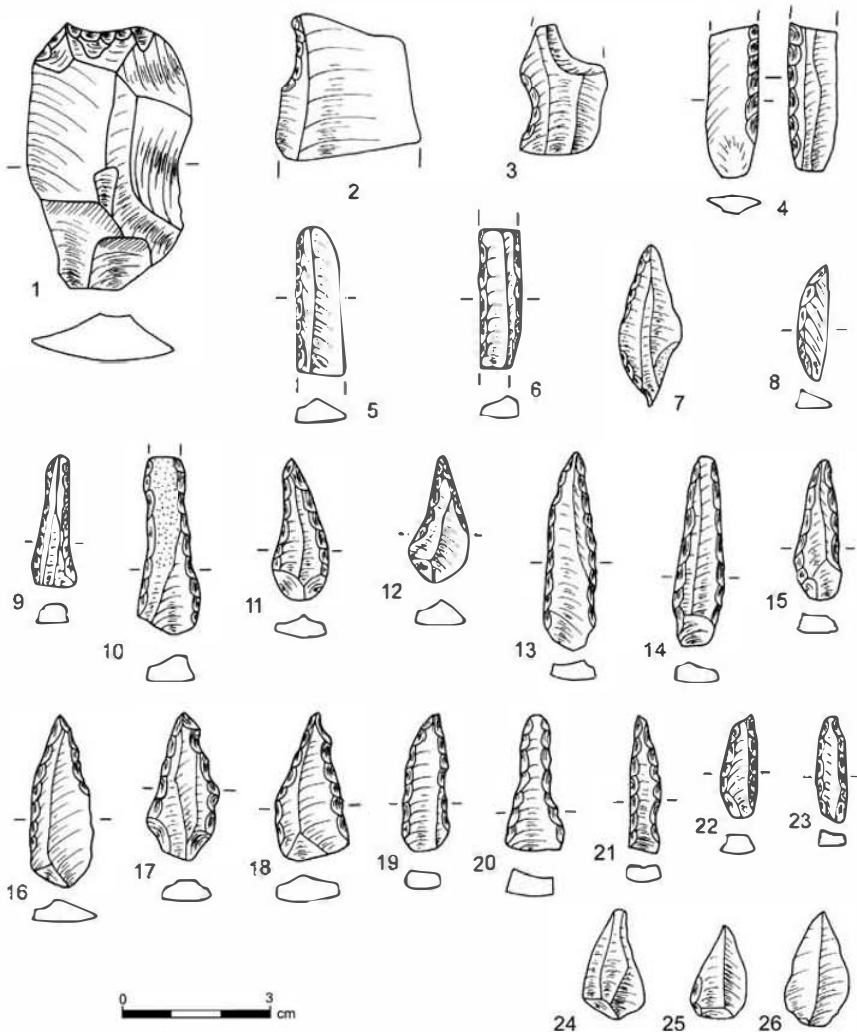


Fig. A4. Huzuq Musa: 1-3-Notches; 4-Bladelet with Helwan retouch; 5-7-Retouched bladelets; 8-Lunate; 9-23-Borers; 24-26-Micro points.

harvesting reeds (Winter 1994), and these grow in abundance on the river banks in the region. Experiments proved that a plain, not serrated, cutting edge is most efficient for cutting reeds (Winter 1994: 271); and actually both at Huzuq Musa and Nahal 'Ein Gev II the cutting edges of most sickles were plain. The absence of arrowheads raises the question of how animal protein was obtained for the menu. Trapping could be a possibility. The function of the abundant borers is not yet known, but perhaps these served for the prepa-

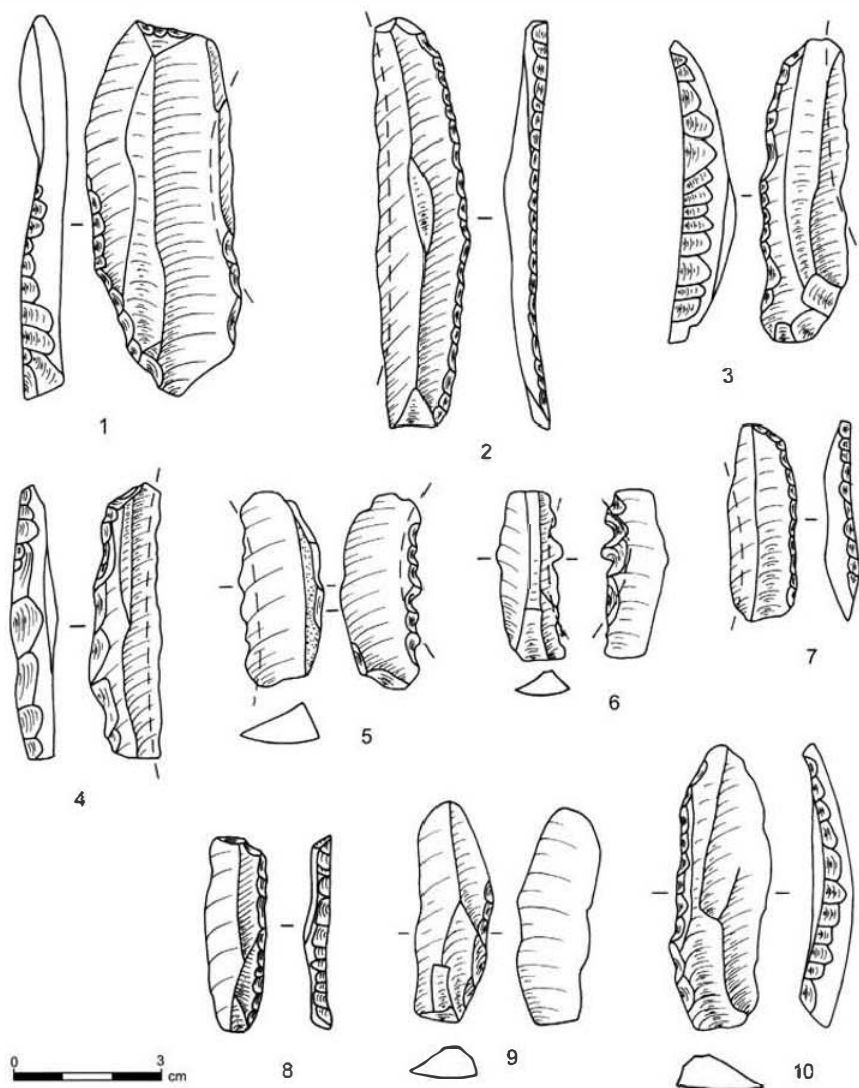


Fig. A5. Huzuq Musa: 1-4, 7-Sickle blades with abrupt retouched back; 5, 6-Denticulated sickle blades; 8-10-Blades with abrupt retouched back.

ration of traps made of reeds. This possibility was already considered at the excavation of Hatula (Ronen et al. 1994: 191). There also an abundance of fish and waterfowl bones was found. The scarcity of similar sites and excavations and the missing data of bone finds make it difficult to draw firm conclusions.

A comparison of surface collections and sieved assemblages is problematic, especially as far as small items are concerned, but in order to demonstrate the similarities from other aspects mentioned above we also present the figures of Nahal 'Ein Gev II in the tables.

A new research project at the site began recently (Rosenberg et al. 2010; Nadel and Rosenberg 2013) that will produce additional important data about this intriguing site.

Period: Epipalaeolithic (final stage) – highly reasonable identification.

Site 48. en-Na'ajeh (2)

Israel Grid 2004/1773, Elevation 240 m b.s.l.

Two flint artefacts without retouch: Chunk 1; Blade 1.

40% of the pottery originate from IA I, and the rest from later periods. The blade has a trapezoidal cross-section similar to Canaan blades, with use wear on one side. This kind of blade is not characteristic of these periods.

Period: IA I – possible identification.

Site 49. en-Na'ajeh (3)

Israel Grid 2003/1772, Elevation 240 m b.s.l.; 5 flint artefacts.

Artefacts without retouch: Primary flake 1; Blade 1.

Tools and retouched artefacts: Retouched flake 1; Notch 2.

80% of the pottery were from IA II. The number of flint items is small, but they fit well into the group of 'ad hoc' tools common in this period. No sickle blades were present. The presence of flint artefacts confirms the use of these at least until IA II.

Period: IA II – reasonable identification.

Site 51. en-Na'ajeh (4)

Israel Grid 2008/1763, Elevation 236 m b.s.l.; 3 flint artefacts.

Artefacts without retouch: Blade 2.

Tools and retouched artefacts: Transversal scraper 1.

30% of the pottery were from the IA II. There is no evidence of earlier occupations; thus it seems reasonable to attribute the flint to this period.

Period: IA II – reasonable identification.

Site 56. Argaman (2)

Israel Grid 1996/1763, Elevation 180 m b.s.l.; 4 flint artefacts (Fig. A6).

Artefacts without retouch: Blade (fragment) 3.

Tools and retouched artefacts: Retouched blade 1.

Some of the artefacts show definite use-wear. All the pottery is attributed to the IA; hence the flint probably also originates from this period.

Period: IA (general) – reasonable identification.

Site 57. Argaman (3)

Israel Grid 1998/1759, Elevation 208 m b.s.l.; 58 flint artefacts (Fig. A6).

Artefacts without retouch: Bladelet core 1; Primary flake 12; Chunk 3; Chip 2; Flake 15; Blade 16; Bladelet 1. Total 50.

Tools and retouched artefacts: Side-scraper 1; Burin 3; Retouched flake 1; Knife with natural back 1; Notch 1; Truncation 1. Total 8.

No diagnostic artefacts were found. The assemblage included a bladelet core, numerous blades with punctiform bases and burins; hence it can be attributed to the Upper Palaeolithic. 50% of the sherds are from IA II, thus some artefacts may originate from this period.

Periods: Upper Palaeolithic – reasonable identification.

IA II – possible identification.

Site 59. Wadi Umm Kharubeh (1)

Israel Grid 1989/1748, Elevation 240 m b.s.l.; 13 flint artefacts (Fig. A6).

Artefacts without retouch: Core 1; Primary flake 3; Flake 5. Total 9.

Tools and retouched artefacts: Retouched flake 1; Backed knife 1; Notch 1; Varia 1. Total 4.

Most of the pottery was from the Late Roman and Byzantine periods and 10% were from IA II. During the IA II, flint tools were mainly improvised for daily use ('ad hoc' tools, Rosen 1997: 34, 158-159). Diagnostic artefacts from prehistoric periods were not found. At the elevation of 240 m b.s.l. it is usually impossible to find prehistoric remnants because during these periods the area

was either below the lake level, or after the rise of the level the remnants were buried by the sediments which accumulated (Bar Tov et al. 2002: Fig. 7). The assemblage, despite the small number of items, includes most components of a local flint industry – the core, primary flakes, flakes and retouched artefacts.

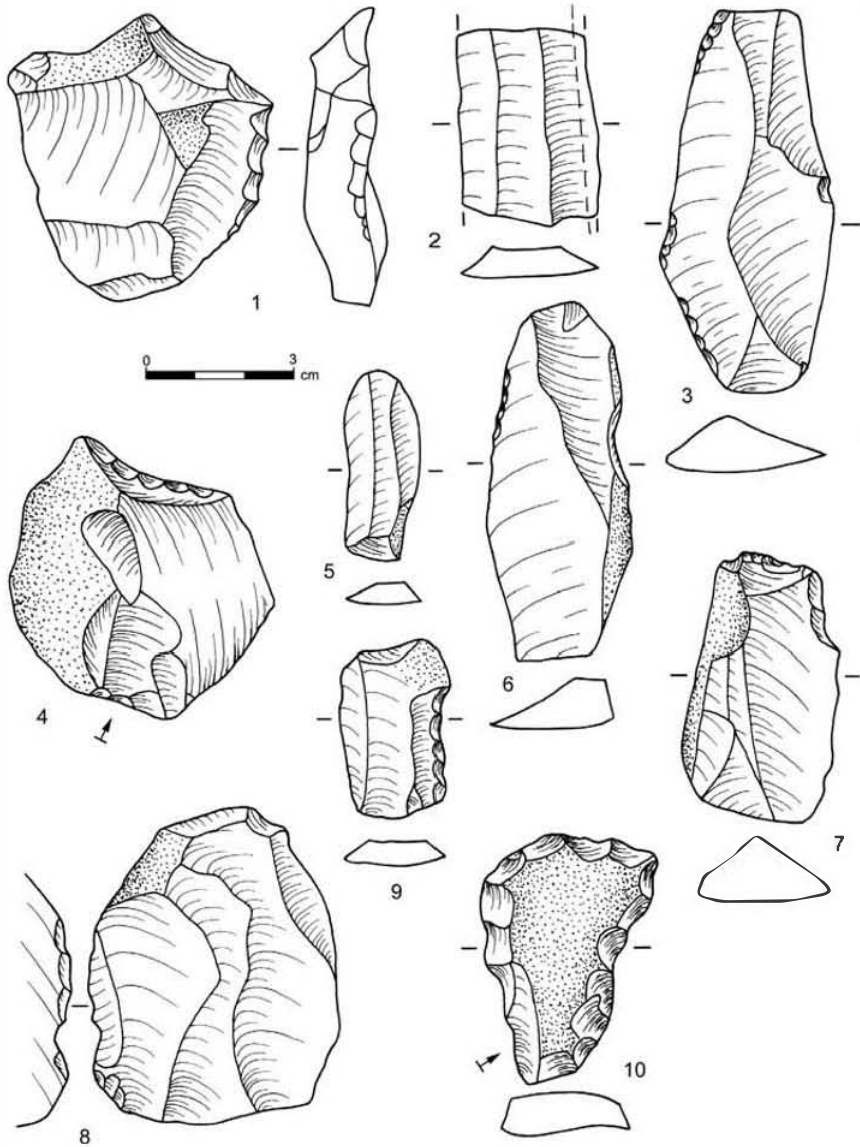


Fig. A6. Kh. Wadi Sadd el-Balqawi: 1-Side-scraper; 2-Canaanite sickle blade. Argaman (2): 3-Retouched blade. Wadi Umm Kharubeh (1): 4-Notch; 7-Naturally backed knife; 8-Retouched flake. Argaman (3): 6-Naturally backed knife. Bedhat esh-Sha'ab: 9-Retouched flake; 10-Side-scraper.

The absence of sickle blades, a most dominant tool during the IA, is remarkable (see Discussion). The reason could be that these pieces were overlooked during the collection, but the phenomenon was observed repeatedly at other contemporary sites in the area, a fact which could point to the character of the society during the IA in this area.

Period: IA II – reasonable identification.

Site 60. Zbeidat

Israel Grid 1995/1748, Elevation 220 m b.s.l.; 17 flint artefacts.

Artefacts without retouch: Flake 10; Blade 3.

Tools and retouched artefacts: Awl 2; Notch 1; Denticulate 1.

The pottery finds are most varied. As no diagnostic artefacts were found the flint assemblage could not be dated.

Period: Not defined.

Site 63. es-Sahel (2)

Israel Grid 1989/1747, Elevation 250 m b.s.l.; 10 flint artefacts.

Artefacts without retouch: Flake 7; Blade 1. Total 8.

Tools and retouched artefacts: Retouched flake 2.

25% of the ceramics were from IA I, 20% from IA II, and the rest from later periods. Diagnostic artefacts from prehistoric periods were not found. Although the assemblage is very small, the bases and bulbs of percussion indicate that the finds were man-made. Here too the absence of sickle blades is remarkable: they are usually found at sites based on a cereal-growing economy. Due to the difficulty in distinguishing between flints from IA I and IA II, the collection was attributed to the IA in general.

Period: IA (general) – highly reasonable identification.

Site 64. es-Sahel (1)

Israel Grid 1986/1746, Elevation 250 m b.s.l.

Four flint artefacts without retouch: Core 1; Primary flake 1; Flake 1; Blade 1.

The assemblage is very small, but the core and the other artefacts indicate that these were purposely flaked. 10% of the sherds were from IA I, 60% from IA II, and the rest from later periods. Lacking any evidence for earlier periods, the flint artefacts should be attributed to the IA, especially IA II.

Periods: IA II – reasonable identification.

IA I – possible identification (part of the collection).

Site 65. Bedhat esh-Sha'ab

Israel Grid 1985/1742, Elevation 240 m b.s.l. 5 flint artefacts (Fig. A6. See also Appendix B).

Artefacts without retouch: Core 1; Primary flake 1; Flake 1.

Tools and retouched artefacts: Side-scraper 1; Retouched flake 1.

Periods according to the excavation: IA I – highly reasonable identification.

IA II – reasonable identification.

Site 66. Siret el-Ma'azeb (2)

Israel Grid 1987/1737, Elevation 260 m b.s.l.

Two flint artefacts without retouch: Core 1; Flake 1.

The small pyramidal core could apparently indicate the existence of an Epipalaeolithic industry. The location of the site at an elevation of 260 m b.s.l. was above the lake level 20,000 years BP (Bar Tov et al. 2002: fig. 7) and artefacts from this period can be found on the surface. We refrain from relying on a single diagnostic artifact - thus the site cannot be dated according to the flint. Period: Not defined.

Site 67. Siret el-Ma'azeb(1)

Israel Grid 1984/1733, Elevation 250 m b.s.l.

Artefacts without retouch: Blade 1.

Tools and retouched artefacts: Backed blade 1.

Both pieces were intentionally flaked artefacts. 45% of the sherds are from IA II, and no indications for earlier periods were found.

Period: IA II – possible identification.

Site 68. Tellul el-Bedha (2)

Israel Grid 1975/1728, Elevation 70 m b.s.l.

Four flint artefacts without retouch: Flake 3; Blade 1.

Only a few artefacts were found, but the bases and the bulbs of percussion

testify to intentional flaking, 15% of the sherds were from the IA, and as no flint artefacts from other periods were found it is reasonable to attribute the artefacts to this period. It should also be noted that similarly to other IA sites in the area, no sickle blades were found.

Period: IA (general) – reasonable identification.

Site 71. Tellul el-Bedha (1)

Israel Grid 1981/1721, Elevation 250 m b.s.l.; 155 flint artefacts (Fig. A7).

Artefacts without retouch: Core 3; Primary flake 15; Chunk 20; Flake 69; Blade 4; Bladelet 3. Total 114.

Tools and retouched artefacts: End-scraper 3; Side-scraper 4; Burin 4; Awl 4; Borer 3; Retouched flake 2; Retouched blade 2; Backed knife 1; Notch 11; Denticulate 1; Truncation 2; Varia 4. Total 41.

Relatively to other sites of the survey this flint assemblage is quite large. Diagnostic artefacts indicating a particular prehistoric period were not found. The products were from light brown to light grey medium-grained flint with some light coloured patina. The assemblage included most components of a local flint industry - cores, primary flakes and waste, intentionally detached flakes and blades and retouched tools. Many of the flakes were rather small, and these, with the three bladelets, could indicate an Epipalaeolithic industry, but without the presence of bladelet cores and retouched microliths it is difficult to support this assumption. 43% of the sherds were from IA I, 45% from IA II, and 5% from the IBA. Without the possibility of distinguishing between flint of IA I and IA II, the bulk of the assemblage is attributed to the IA generally. Here also the absence of sickle blades, a dominant flint tool of the IA, is significant.

Period: IA (general, the bulk of the assemblage) reasonable identification.

Site 72. el-Makhruq (1)

Israel Grid 1984/1715, Elevation 250 m b.s.l.

Only a single axe (broken) was found. 60% of the sherds were from Bronze and Iron Ages, and the axe is not characteristic in these periods. The cross-section of the tool is lentoid, which is typical mainly of the Neolithic period. The absence of more flint artefacts raises questions about the thoroughness of the collecting. The identification of the tool is correct, but a single item is not enough to indicate a Neolithic presence.

Period: Not defined.

Site 73. el-Makhruq (2)

Israel Grid 1984/1709, Elevation 240 m b.s.l., 12 flint artefacts (Fig. A7).

Artefacts without retouch: Chip 2; Flake 2; Bladelet 1. Total 5.

Tools and retouched artefacts: Borer 2; Retouched flake 1; Backed blade 1;

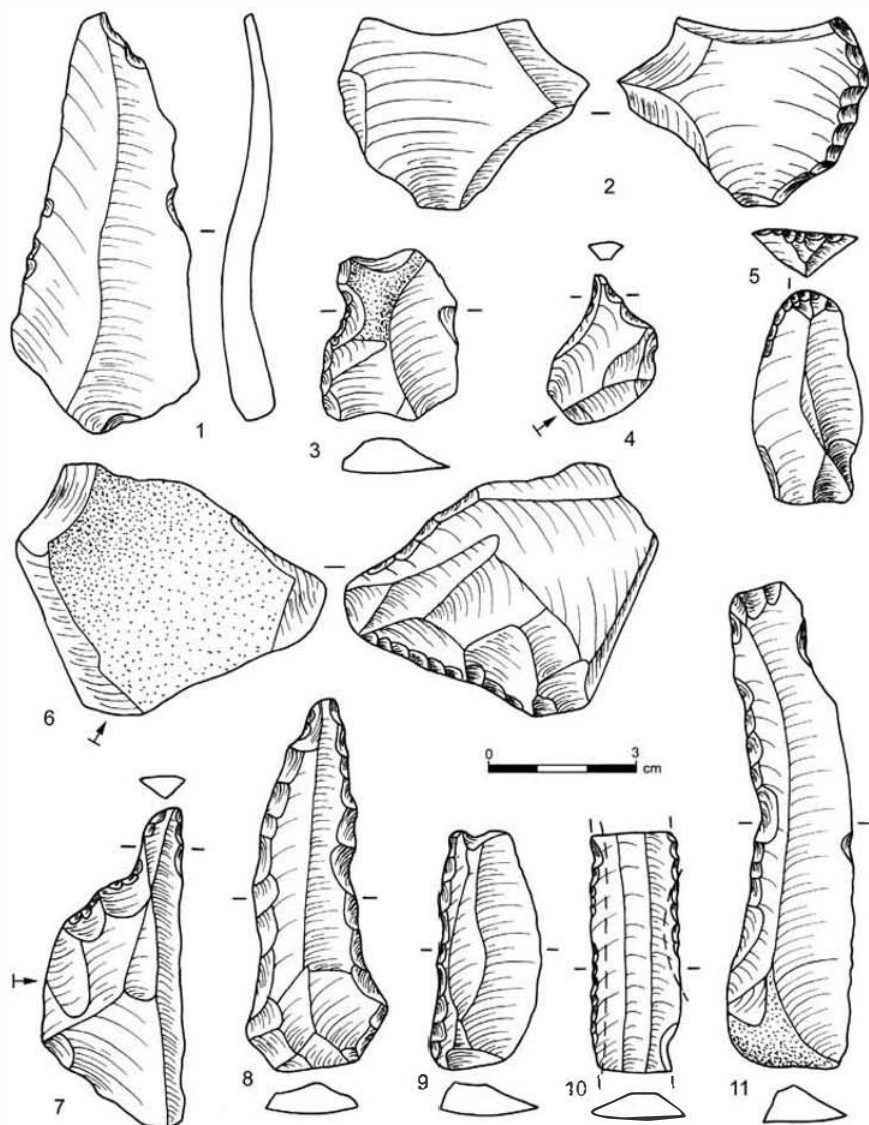


Fig. A7. Tellul el-Bedha (1): 1-Blade; 2, 6-Side-scrapers; 3-Notch; 4-Awl; 5-End-scraper; 7-Borer. el-Makhruq (2): 8-Borer; 9-Backed blade; 10-Canaanean sickle blade; 11-Backed knife.

Backed knife 1; Canaanean sickle blade 1; Varia 1. Total 7.

71% of the sherds were from IA II, 23% from EBA, and 5% from MBA. Most flint artefacts are of light brown, uniform flint. The single diagnostic tool is the Canaanean sickle blade attributed to the EBA and IBA. The assemblage could be mixed, but no diagnostic tool of the IA (e.g. geometric sickles) was found.

Periods: EBA – highly reasonable identification (part of the assemblage).

IA II – reasonable identification (part of the assemblage).

Summary of Landscape Unit 28

A total of 2534 flint artefacts were collected at 20 sites in Landscape Unit 28. For three sites the period could not be identified due to the scarcity of artefacts and the lack of diagnostic pieces. No flint artefacts from the Middle Palaeolithic period were found: these, if present, would have been buried under the deposits of Lake Lisan (Bar Tov et al. 2002: fig. 7).

Site 47 (Huzuq Musa) from the terminal Epipalaeolithic is of special importance, as only one contemporary site from this stage has been discovered to date – Nahal 'Ein Gev II (Bar Yosef and Belfer-Cohen 2000).

The absence of flint artefacts from the Chalcolithic, MBA and LBA is significant.

One EBA site and one IBA site were identified as highly reasonable.

Only a few IA I sites were identified as reasonable and highly reasonable, but from IA II the number increases. There is no way to distinguish between flints from these periods, and the identification was mainly according to the pottery. The importance of these flint artefacts is their mere presence in many sites of these stages, thus confirming the continuity of the production and utilization of flint as an integral part of the way of life in the IA. It should be mentioned that not even one sickle blade – a most important tool in the assemblages of the period – was found among the artefacts from the IA. Their absence here could seemingly be caused by insufficient collecting, but their absence from all IA sites is consistent and provides evidence about the character of the sites. All the sites were in a semi-arid area, close to the River Jordan. The absence of sickle blades proves that the population was not engaged in extensive farming. There is a possibility that insufficient precipitation was the reason, but it seems more reasonable that the population continued the traditional way of life as nomadic or semi-nomadic herders. In this region during the IA, the economy was apparently based on traditional herding, either because of local conditions, or due to both these factors.

Landscape Unit 28. Summary of sites by period. Identification according to flint artefacts			
Period	Highly reasonable	Reasonable	Possible
Middle Palaeolithic	-	-	-
Upper Palaeolithic	-	1	-
Epipalaeolithic	1	-	-
Neolithic	-	-	-
Chalcolithic	-	-	-
Early Bronze (EBA)	1	-	-
Intermediate Bronze (IBA)	1	-	-
Middle Bronze (MBA)	-	-	-
Late Bronze (LBA)	-	-	-
Bronze general (BA)	-	-	-
Iron I (IA I)	1	1	2
Iron II (IA II)	-	6	4
Iron general (IA)	1	3	-
Not defined	3		

Table A4: Landscape Unit 28 - Periods and identification level.

Landscape Unit 29

Site 75. Khirbet Wadi Umm Watad (1)

Israel Grid 1949/1751, Elevation 160 m b.s.l. Two blades without retouch.

80% of the sherds originate from the EBA I, and 20% from the EBA II. The flint artefacts belong to one of these periods, but it is not known which.

Period: EBA – highly reasonable identification.

Site 76. Khirbet Wadi Umm Watad (2)

Israel Grid 1954/1754, Elevation 50 m b.s.l.; 8 flint artefacts.

Artefacts without retouch: Core 1; Chunk 1; Blade 1. Total 3.

Tools and retouched artefacts: Burin 1; Awl 1; Retouched blade 2; Notch 1. Total 5.

60% of the pottery were from the IA. The awl was prepared from a flake with

a faceted base. This technique is common in the Middle Palaeolithic, and sites from this period were found not far away (Landscape Unit 30 below, see also Winter 1996: 670-676). It may be that artefacts from this period were collected for reuse and retouched again, similarly to the IA site el-Ahwat (Winter 2012: 342-345). The new retouch without patina which is clearly visible on some artefacts confirms this assumption. As no further evidence for the Middle Palaeolithic was found, the assemblage was attributed to the IA. Here also no sickle blades were found.

Period: IA (general) – highly reasonable.

Site 81. Khallet el-Fuleh

Israel Grid 1945/1735, Elevation 200 m b.s.l.; 14 flint artefacts (Fig. A8).

Artefacts without retouch: Core 1; Primary flake 1; Flake 3; Blade 2. Total 7.

Tools and retouched artefacts: Fan-scraper 1; Side-scraper 1; Awl 1; Retouched flake 1; Naturally backed knife 1; Notch 1; Side-scraper/Borer 1. Total 7.

Diagnostic tools were not found, but all the pottery sherds were IBA, and the flint assemblage was attributed similarly.

Period: IBA – highly reasonable.

Site 82. Tell esh-Sheikh Mazar

Israel Grid 1959/1719, Elevation 230 m b.s.l.; 9 flint artefacts.

Artefacts without retouch: Primary flake 1; Flake 2; Blade 2. Total 5.

Tools and retouched artefacts: End-scraper 1; Side-scraper 1; Retouched flake 2. Total 4.

A few sherds were from EBA and IA, but missing diagnostic flints did not allow the dating of the collection, even though it should be noted that the artefacts were knapped and retouched. The artefacts could have originated from both periods.

Period: Not defined.

Site 83. Wadi Umm Hajar (1)

Israel Grid 1948/1708, Elevation 190 m b.s.l.; 33 flint artefacts (Fig. A8).

Artefacts without retouch: Core 1; Primary flake 4; Flake 10; Blade 4. Hammerstone 3. Total 22.

Tools and retouched artefacts: End-scraper 1; Transversal scraper 2; Burin 2;

Awl 1; Retouched blade 1; Notch 2; Truncation 1; Varia 1. Total 11.

The collection presents the full sequence of a lithic industry: the core, primary flakes, the hammerstones (two broken), blades, flakes and retouched tools. Bases and bulbs of percussion indicate intentional flaking. Diagnostic artefacts were not found, and for the preparation of most tools only minimal

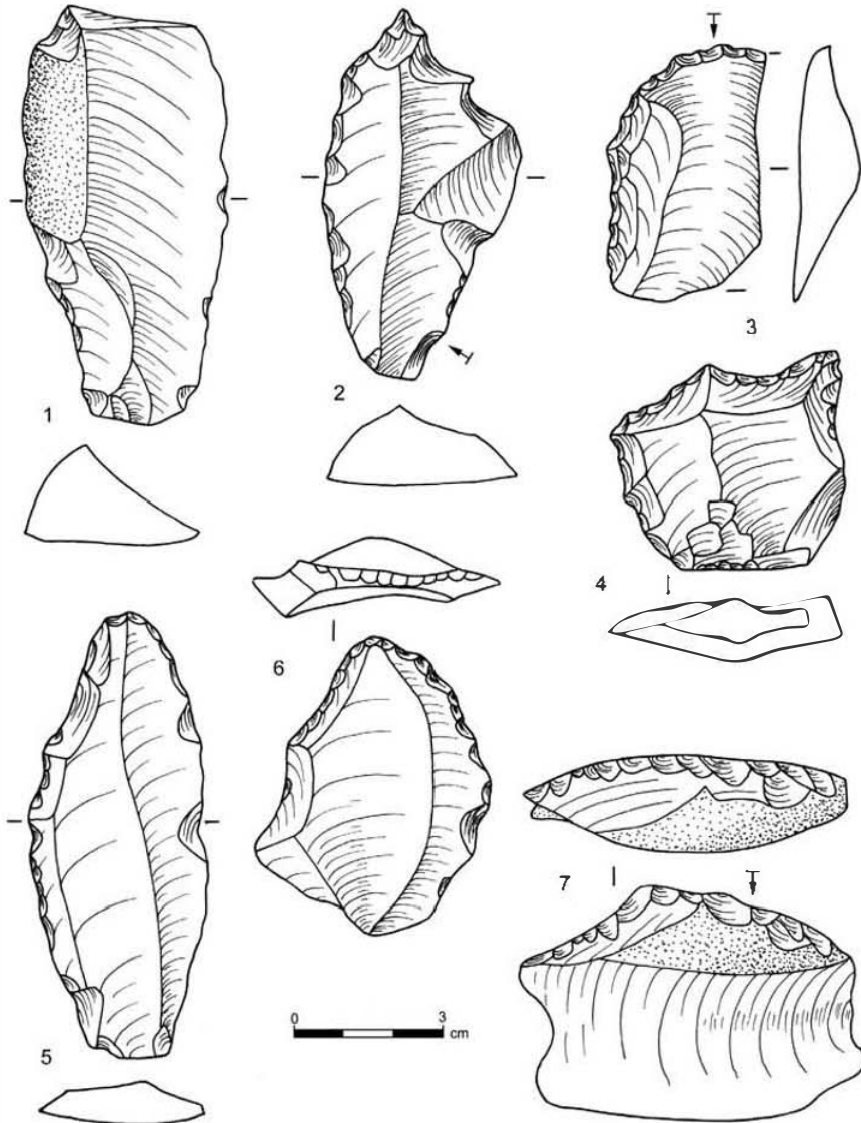


Fig. A8. Khallet el-Fuleh: 1-Naturally backed knife; 2-Awl; 3-Fan scraper (broken); 4-Side-scraper/borer. Wadi Umm Hajar (1): 5-Retouched blade; 6-End-scraper; 7-Transversal scraper (ventral retouch).

effort was invested. These tools belong to the category of 'ad hoc' improvised tools characteristic of historical times (Rosen 1997: 34, 158-159). 56% of the pottery finds are from IA I, 16% from IA II, and the rest from later periods. Apparently the bulk of the flint assemblage is from IA I, despite the fact that at this stage the difference between flint products of IA I and IA II cannot be distinguished. At this site also, even though it is located uphill, no typical sickle blades were found.

Periods: IA I – highly reasonable identification.

IA II – reasonable identification.

Site 86. Rujm es-Si'a

Israel Grid 1956/1693, Elevation 200 m b.s.l.; 87 flint artefacts (Fig. A9).

The site was first surveyed in 1993 and in 2002 it was surveyed again.

Most flint artefacts were found on a slope at the foot of a rock shelter, elevation about 205 m to 215 m b.s.l., a short distance north of the fortress. 15% of the pottery were from IA.

1993 survey: 69 flint artefacts

Artefacts without retouch: Flake core 4; Blade core 1; Primary flake 8; Flake 6; Blade 3; Blade fragment 3; Bladelet fragment 2; Chunk 4. Total 31.

Tools and retouched artefacts: End-scraper on flake 1; End-scraper on blade 1; Déjeté scraper on blade 1; Carinated, nosed end-scraper 1; Concave side-scraper 2; Convex side-scraper on primary flake 1; Angled burin 1; Dihedral burin on primary flake 1; Dihedral burin on blade 1; Burin on distal notch 2; Burin on bladelet core 1; A-typical burin 1; End-notch on flake 3; Notch on flake 3; Notch on blade 4; Double notch on flake 1; Awl on flake 1; Retouched flake 3; Retouched blade 2; Backed knife 1; Truncated blade 1; Emireh point 1; Varia 3. Total 38.

2002 survey: 18 flint artefacts

Artefacts without retouch: Primary flake 1; Flake 4; Blade 4. Total 9.

Tools and retouched artefacts: Side-scraper 2; Burin 2; Awl 1; Retouched blade 1; Notch 3. Total 9.

The site is located near Wadi Far'ah, on a moderate slope at the foot of two caves (with recent occupational activities), close to a rock shelter. The survey did not include the caves, but was carried out on the east-facing slope. The geographical location is similar to other prehistoric sites from the Upper Palaeolithic, such as Wadi Fazaal about 10 km to the south, at elevations of about 170 m b.s.l. (Goring-Morris 1980a). Apparently these sites were located similarly to Rujm es-Si'a, close to the ancient western shoreline of Lake Lisan near the mouth of a river flowing from west to east.

The investigation of the 1993 collection indicated that the assemblage consisted of two different parts. The first were abraded and rolled artefacts from grey to dark violet flint with a heavy white or brown patina. In this group no tools were found, only a few pieces with marginal secondary retouch. The second part consisted of brown pieces without patina, and grey to violet pieces with a light spotted white to grey patina. This group was not abraded or

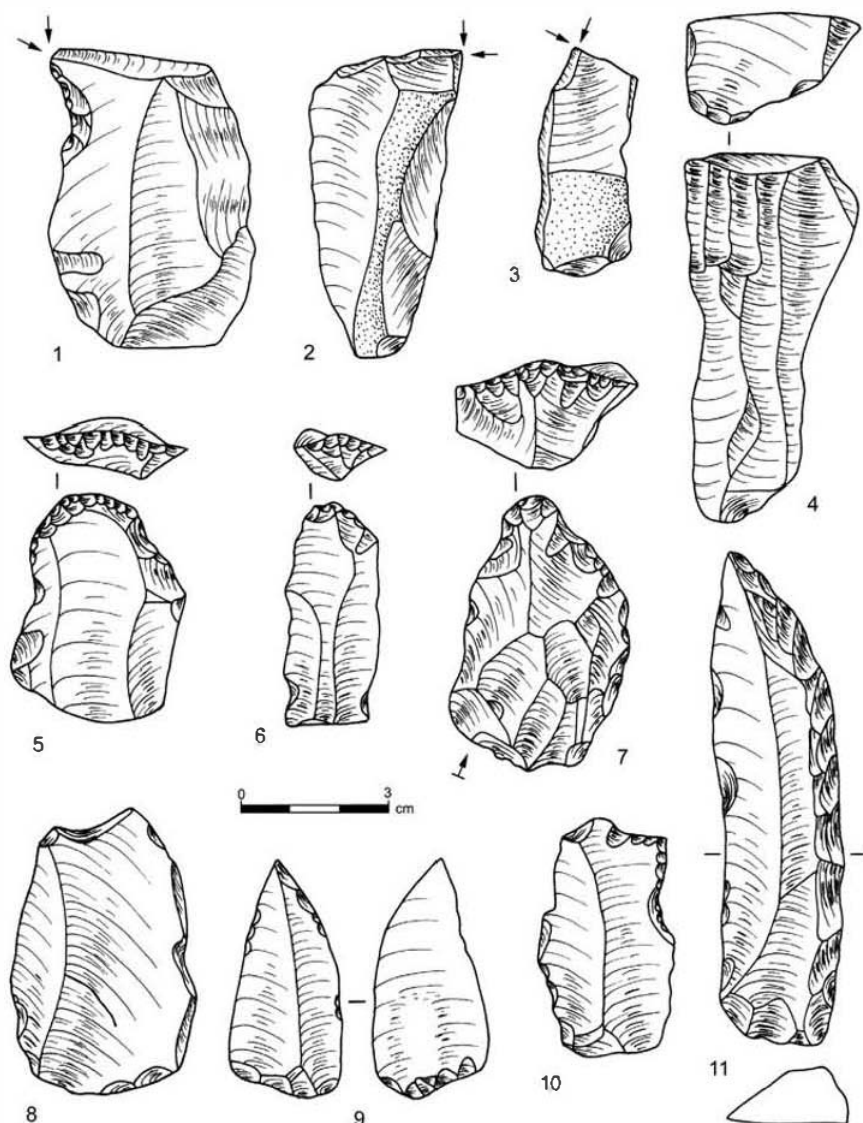


Fig. 9. Rujm es-Si'a: 1-3-Burins; 4-Core; 5-End-scraper on flake; 6-End-scraper on blade; 7-Carinated, nosed-end-scraper; 8-Distal notch; 9-Emireh point; 10-Awl; 11-Backedknife.

rolled, the pieces were sharp, and at least some were unearthed only recently, perhaps by modern digging. The elevation (about 200 m b.s.l.) indicates that the site was flooded by the lake for some time (Bar Tov et al. 2002: fig. 7). The layer of the lake sediments covering the artefacts was apparently relatively thin; thus the flints were easily exposed.

The modest number of diagnostic artefacts does not prevent chronological conclusions based on the composition of the assemblage. The dominant items were notches (29.8%), burins (19.1%), and various scrapers (8.5%). Similar tool compositions characterize assemblages from the Upper Palaeolithic, perhaps from the 'Levantine Aurignacian' stage (Gilead 1981; Marks 1981; Kaufman 1988). In these assemblages flakes, burins, and scrapers are dominant, and the share of microliths and bladelets is rather low. Some researchers assume that the picture is more complicated, and that more stages should be considered, but this issue is beyond the scope of this study. The assemblage from our site can be compared to Fazeel IX, about 10 km to the south. There also the percentage of bladelets and microliths is rather low, burins constitute an important part, and notches and scrapers are present (Goring-Morris 1980a: 178). Most burins are angled or on a notch, and some notches are distal without retouch ('Clactonian', Goring-Morris 1980a); these phenomena were also observed in our assemblage. The cores and the knapped items point to a flake-dominated industry similar to Fazeel IX. The thick convex abrupt-backed blade is quite rare in this period, but present at some sites: 'Ein Aqev D31 (Marks 1976: 106 and fig. 4-16j) and 'Ein Aqev East D34 (Ferring 1977: 255 and fig. 9-19a). The burin on a bladelet core is further evidence for attributing the main part of the assemblage to this period. Another noteworthy item is the Emireh Point. This artefact is usually attributed to an earlier period, the transition to the early Upper Palaeolithic, for example Lower Boker (Marks and Kaufman 1983: table 5-17 and fig. 5-24h), but no conclusions can be drawn from a single artefact.

Another site attributed to the same stage of the period is Urqan es-Safa, Site 185, but this site is located higher, and was never flooded by the lake. Possibly a part of the assemblage should be attributed to IA.

Periods: Upper Palaeolithic – highly reasonable identification (main part of the finds).

IA II (general) – possible identification (part of the finds).

Site 89. Masu'a (1)

Israel Grid 1962/1675, Elevation 200 m b.s.l.; 93 flint artefacts (Fig. A10).

Artefacts without retouch: Core 3; Primary flake 22; Core trimming element 1; Chunk 4; Chip 2; Flake 25; Blade 4. Total 61.

Tools and retouched artefacts: End-scraper 4; Fan-scraper 1; Side-scraper 2; Burin 1; Borer 2; Retouched flake 3; Retouched blade 4; Notch 9; Denticulate 2; Axe (unfinished) 1; Varia 3. Total 32.

The assemblage is rather large compared to other sites of the survey. It contains all the components of a regular production process – cores and waste, flakes, blades and retouched tools. Notches constitute 28% of the tool category, perhaps indicating that some special activity was carried out here. Most tools were large and crude, and only minimal effort was invested in their production. No sickle blades or any other diagnostic artefacts (except the fan-scraper) indicating other prehistoric periods were found. The pottery collection was attributed to the Chalcolithic; thus the flint collection should be attributed similarly.

Period: Chalcolithic – highly reasonable identification.

Site 90. Masu'a (5)

Israel Grid 1966/1673, Elevation 255 m b.s.l.; 28 flint artefacts (Fig. A10).

Artefacts without retouch: Core 1; Primary flake 6; Chunk 3; Flake 6; Blade 3. Total 19.

Tools and retouched artefacts: End-scraper 1; Side-scraper 1; Retouched blade 1; Notch 6. Total 9.

Here also no diagnostic artefacts were found, and all sherds range from Byzantine times to the Middle Ages. The finds resemble those from the previous site (Site 89 Masu'a [1]) and the next site (Site 91 Masu'a [2]), and indicate a regular industrial process. Similarly to the previous site, most artefacts are crude, the share of notches relatively high, and no sickle elements were found. The similarity of the assemblages and the proximity of the sites indicate that the sites were contemporary.

Period: Chalcolithic – possible identification.

Site 91. Masu'a (2)

Israel Grid 1961/1671, Elevation 200 m b.s.l.; 85 flint artefacts (Fig. A11).

Artefacts without retouch: Core 3; Primary flake 13; Flake 29; Blade 17; Bladelet 1. Total 63.

Tools and retouched artefacts: Déjeté scraper 1; Transversal scraper 1; Burin 2; Borer 4; Retouched flake 1; Backed knife 1; Notch 5; Denticulate 2; Truncation 3; Varia 2. Total 22.

The flint assemblage attests a full production sequence, but no diagnostic

items were found. Again here the relative share of notches is high, perhaps indicating some special activity, and no sickle elements were present. The pottery collection is attributed to the Chalcolithic, and so is the flint collection. Period: Chalcolithic – highly reasonable identification.

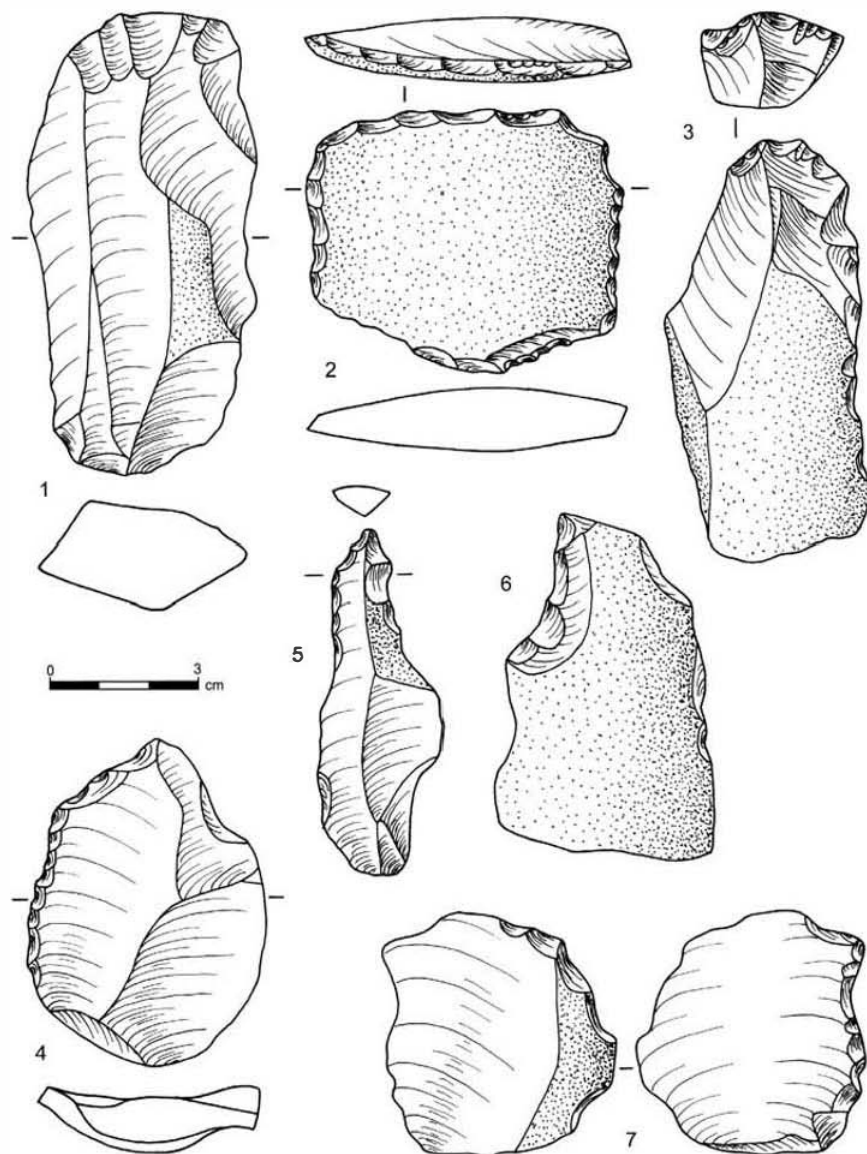


Fig. A10. Masu'a (1): 1-Axe; 2-Fan-scraper; 3-End-scraper; 4-Side-scraper; 5-Borer; 6-Notch. Masu'a(5) : 7-Side-scraper (ventral retouch).

Site 92. Masu'a (4)

Israel Grid 1964/1668, Elevation 230 m b.s.l.; 3 flint artefacts (Fig. An).

Artefacts without retouch: Primary blade 1; Flake 1.

Tools and retouched artefacts: Backed blade 1.

The few items found were definitely man-made, but no diagnostic artefacts were present. All pottery was from IA I; thus the flint items were also attributed to this period. Again it has to be remarked that no sickle blades were found at this site.

Period: IA I – reasonable identification.

Site 93. E.P. -189

Israel Grid 1961/1669, Elevation 200 m b.s.l.; 10 flint artefacts (Fig. An).

Artefacts without retouch: Chunk 3; Flake 2; Blade 1; Bladelet 1. Total 7.

Tools and retouched artefacts: End-scraper 2; Geometric sickle blade 1. Total 3.

24% of the pottery were from IA I, 11% from EBA, and the rest from later periods. A triangular artifact was attributed according to its shape to the Geometric Sickle group, even though no sickle sheen was visible. The attribution to IA I seems reasonable.

Periods: IA I – reasonable identification.

EBA – possible identification.

Site 95. Masu'a (6)²

Israel Grid 1962/1666, Elevation 210 m b.s.l.; 41 flint artefacts.

Artefacts without retouch: Flake 21; Blade 17. Total 38.

Tools and retouched artefacts: Scraper 2; Perforator 1. Total 3.

70% of the pottery were from IA I. No diagnostic flints were found, but in accordance with the larger number of IA I sherds, it is reasonable that the flint is from this period.

Period: IA I – reasonable identification.

² The flint from this site were analysed by Sonia Pinsky.

Site 96. Masu'a (9)

Israel Grid 1959/1666, Elevation 175 m b.s.l.; 19 flint artefacts (Fig. A11).

Artefacts without retouch: Primary flake 2; Flake 5; Blade 6. Total 13.

Tools and retouched artefacts: Side-scraper 1; Awl 1; Retouched flake 1; Notch 1; Denticulate 1. Total 6.

35% of the pottery were Chalcolithic or EBA, 10% IA I, and 5% IA II. No diagnostic flints were found, and the collection could be attributed to any of these periods. In accordance with the larger number of Chalcolithic/EBA sherds it is reasonable that most flint artefacts were from these periods.

Periods: Chalcolithic/EBA – reasonable identification.

IA – possible identification.

Site 97. Masu'a (3)

Israel Grid 1961/1668, Elevation 189 m b.s.l.

Four flint artefacts without retouch: Chunk 1; Flake 2; Bladelet 1.

7% of the pottery were from IA I, and 42% from IA II. The flint artefacts were man-made, but there were no indications of any particular period. It seems reasonable to attribute the flint to the IA.

Periods: IA I – possible identification.

IA II – reasonable identification.

Site 98. Tell 'Abeid (1)

Israel Grid 1968/1662, Elevation 200 m b.s.l.; 35 flint artefacts (Fig. A12).

Artefacts without retouch: Primary flake 3; Chunk 6; Flake 9; Blade 5; Bladelet 2. Total 25.

Tools and retouched artefacts: End-scraper 1; Side-scraper 2; Retouched flake 2; Notch 5. Total 10.

Diagnostic artefacts were not found, but the composition of the flint assemblage indicates a regular industrial process. 6% of the pottery were from IA I, 54% from IA II, and the rest was Roman-Byzantine. In the absence of evidence of previous periods the bulk of the collection is attributed to IA II. Despite the rather large collection no typical sickle blades were found.

Periods: IA I – possible identification.

IA II – reasonable identification.

Site 99. Tell 'Abeid (2)

Israel Grid 1967/1660, Elevation 200 m b.s.l.; 14 flint artefacts (Fig. A12).

Artefacts without retouch: Primary flake 1; Flake 4. Total 5.

Tools and retouched artefacts: Burin 1; Backed knife 2; Notch 5; Varia 1. Total 9.

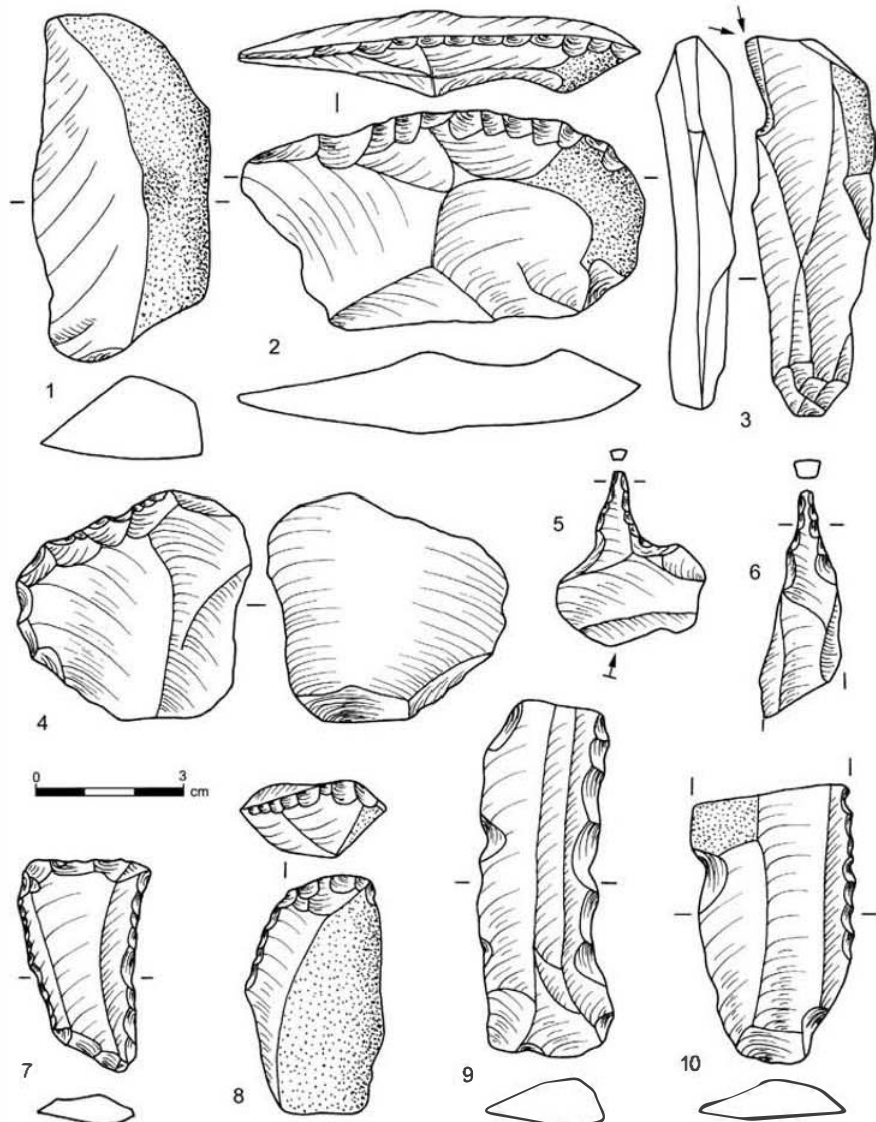


Fig. A11. Masu'a (2): 1-Naturally backed knife; 2-Transversal scraper; 3-Burin; 4-Oblique-scraper; 5,6-Borers. E.P. -189; 7-Geometric sickle blade; 8-End-scraper. Masu'a (4): 9-Backed blade. Masu'a (9): 10-Denticulate.

6% of the pottery were from the MBA, 10% from IA I, 50% from IA II, and the rest from later periods. In the absence of evidence of previous periods the bulk of the collection is attributed to IA II.

Periods: MBA – possible identification (part of the collection).

IA I – possible identification (part of the collection).

IA II – reasonable identification (part of the collection).

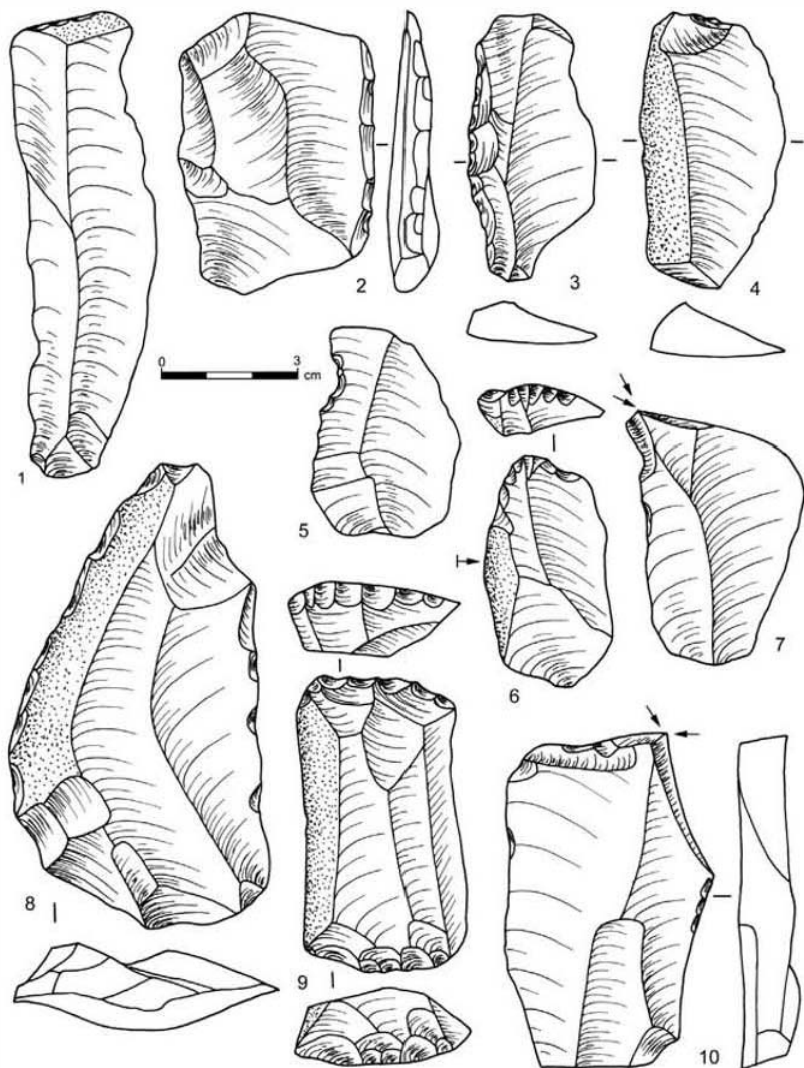


Fig. A12. Tell 'Abeid (1): 1 - Blade; 2 - Side-scraper; 6 - End-scraper. Tell 'Abeid (2): 3 - Backed knife; 4 - Naturally backed knife; 5 - Notch; 7 - Burin. Yafit (4): 8 - Backed knife; 9 - Double end-scraper; 10 - Burin.

Site 100. Yafit (8)

Israel Grid 1961/1650, Elevation 200 m b.s.l.; 75 flint artefacts.

Artefacts without retouch: Primary flake 7; Chunk 8; Flake 37; Blade 16; Ridge-blade 1; Levallois flake 3. Total 72.

Tools and retouched artefacts: End-scraper 1; Naturally backed knife 1; Borer 1. Total 3.

Many artefacts were worn or burnt. 80% of the pottery were from IA II, and most flint artefacts should probably be attributed to this period. The three Levallois flakes with faceted bases are diagnostic for the Middle Palaeolithic, but apparently they do not testify to a human presence here in this period, as since the Middle Palaeolithic the site at this elevation was covered by a thick layer of lake deposits and the remnants of this period are buried. Only artefacts which were swept down from higher regions can be found on the surface. No diagnostic sickle blades were found.

Period: IA II – reasonable identification.

Site 102. Yafit (7)

Israel Grid 1950/1640, Elevation 210 m b.s.l.

One flake without retouch.

Period: Not defined.

Site 104. Yafit (4)

Israel Grid 1941/1631, Elevation 220 m b.s.l.; 84 flint artefacts (Fig. A12).

Artefacts without retouch: Primary flake 8; Chunk 2; Flake 27; Blade 33. Total 70.

Tools and retouched artefacts: Double end-scraper 1; Side-scraper 1; Burin 1; Backed knife 2; Denticulate 7; Truncation 2. Total 14.

No diagnostic artefacts were found, but the composition of the collection points to the Upper Palaeolithic period (approximately 45000–20000 BP), as indicated by the large number of blades and the presence of the burin and the double end-scraper. During the relevant period, from 45000 to 30000 years BP, the lake level varied around 280 m b.s.l. At 20000 years BP the level rose, and then varied between 200 m b.s.l. and 165 m b.s.l. (Bar Tov et al. 2002: fig. 7). Thus it is obvious that the site was flooded for a certain period by the lake, covered by lake deposits, and remnants of the period can barely be found on the surface. Only when these remnants were exposed by natural forces or

human activities they were located on the surface. 15% of the pottery were from IA I, 60% from IA II, and the rest was LR. Although the composition of the assemblage confirms the attribution to the Upper Palaeolithic, some of the artefacts could originate from the IA.

Periods: Upper Palaeolithic – highly reasonable identification.

IA I – possible identification.

IA II – reasonable identification.

Site 105. Yafit (3)

Israel Grid 1939/628, Elevation 230 m b.s.l.; 24 flint artefacts (Figs. A13; A14).

Artefacts without retouch: Core 1; Primary flake 12; Blade 5. Total 18.

Tools and retouched artefacts: End-scraper 1; Rounded scraper 1; Retouched flake 1; Notch 1; Denticulate 1; Chopper 1. Total 6.

The composition of the assemblage indicates a regular knapping process. The artefacts are large and crude, and made from violet to brown flint, with some light-coloured patina. The tools may be attributed to the group defined as 'ad hoc' tools (Rosen 1997: 34, 158-159). 30% of the pottery were from IA I, 18% from IA II, and the rest from later periods. An excavation in February 2005 unearthed mainly IA I sherds, and most of the assemblage should be attributed similarly. Despite the fact that sickle elements are common in IA sites in other areas, no such tools were found here.

Periods: IA I – highly reasonable identification.

IA II – reasonable identification.

Site 106. Yafit (6)

Israel Grid 1937/628, Elevation 190 m b.s.l.; 7 flint artefacts.

Artefacts without retouch: Primary flake 1; Chunk 1; Flake 2. Total 4.

Tools and retouched artefacts: Borer 1; Retouched flake 1; Notch 1. Total 3.

The few artefacts, from flint similar to that of the previous site (Yafit [3]), do not include any diagnostic items. The collection should be attributed to the IA, similarly to the other Yafit sites. There were no indications of earlier periods.

Period: IA (general) – possible identification.

Site 107. Yafit (2)

Israel Grid 1939/1625, Elevation 250 m b.s.l.; 7 flint artefacts.

Artefacts without retouch: Core 2; Primary flake 3. Total 5.

Tools and retouched artefacts: Retouched blade 1; Notch 1. Total 2.

32% of the pottery finds were from IA 1, and the rest from LR. The few artefacts were made from brown to violet flint, with some light-coloured patina, and were extremely large. The site resembles the previous site (Yafit [3]) at nearly the same elevation. As no items from earlier periods were present, the flint finds should also be attributed to IA 1. Again, similarly to most IA sites in this area, no typical sickle blades were found.

Period: IA 1 – reasonable identification.

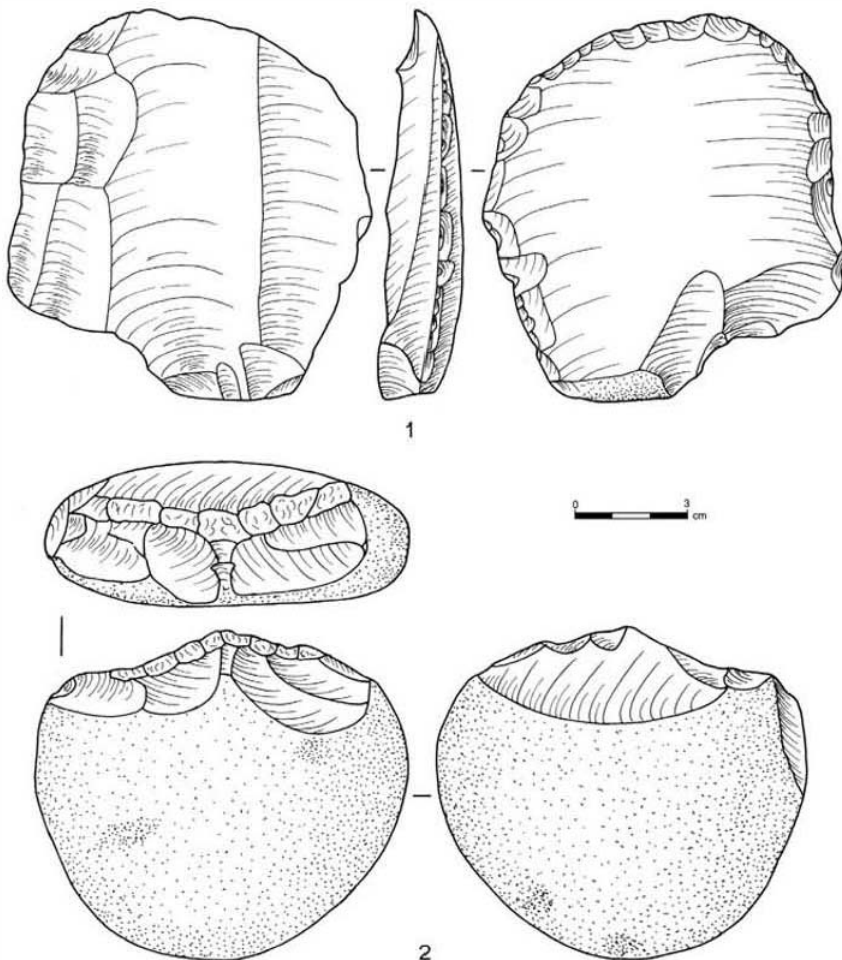


Fig. A13. Yafit (3): 1-Rounded scraper (ventral retouch); 2-Chopper on flint cobble.

Site 108. Yafit (1)

Israel Grid 1939/1623, Elevation 250 m b.s.l.; 18 flint artefacts (Fig. A14).

Artefacts without retouch: Core 1; Primary flake 2; Core trimming element 1; Flake 6; Blade 4. Total 14.

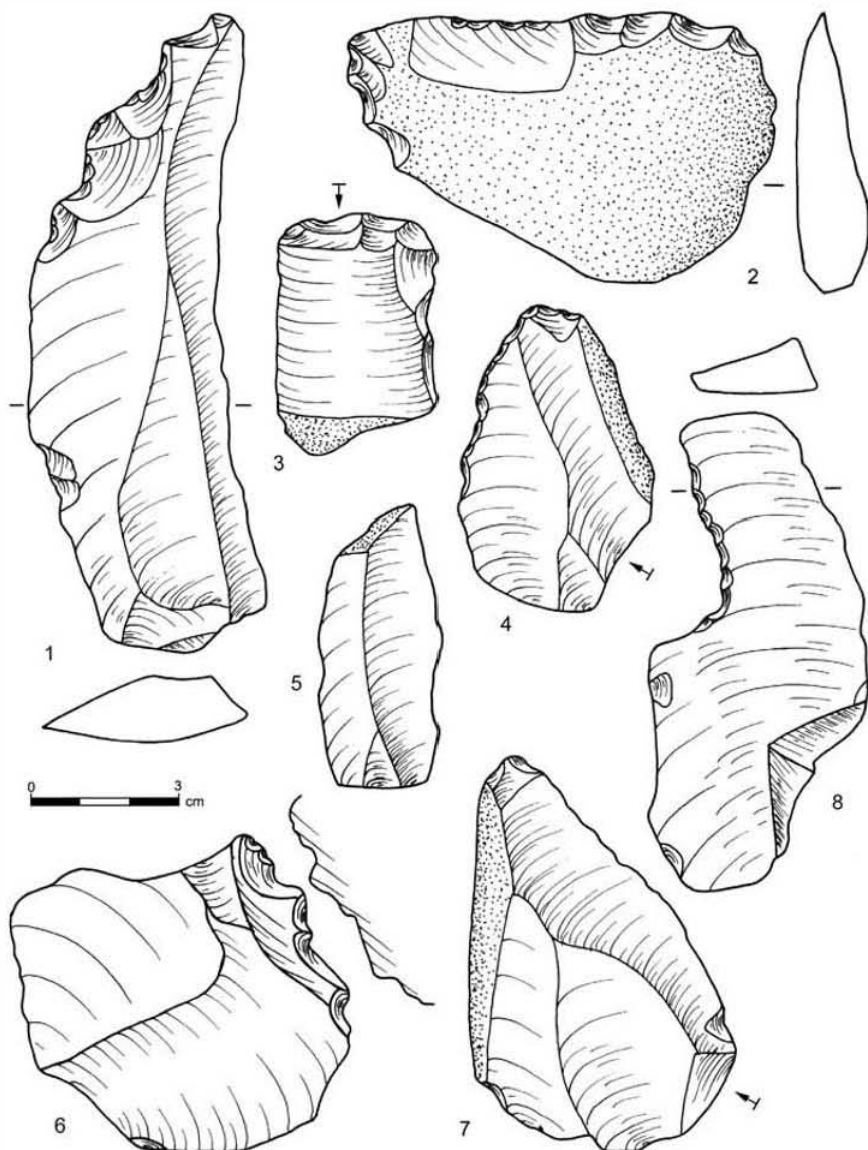


Fig. A14. Yafit (3): 1-Denticulate. Yafit (1): 3-Truncation on ventral face; 4-Retouched flake; 5-Blade. Yafit (5): 2-Transversal knife; 6-Denticulate; 7-Naturally backed knife; 8-Ventral notch.

Tools and retouched artefacts: Retouched flake 1; Notch 1; Truncation 2. Total 4.

The core, core trimming element, primary flakes and all other artefacts indicate a local flint industry. The brown to violet flint variety, with some light-coloured patina resembles the variety of the other Yafit sites, and the tools are from the 'ad hoc' tool group. 37% of the pottery were from IA I, and the rest LR. No items from earlier periods were found; thus the flint finds should possibly be attributed to IA I. Similarly to most IA sites in this area, no typical sickle blades were found.

Period: IA I – possible identification.

Site 109. Yafit (5)

Israel Grid 1936/1626, Elevation 190 m b.s.l.; 27 flint artefacts (Fig. A14).

Artefacts without retouch: Primary flake 5; Chunk 1; Flake 7; Blade 4. Total 17.

Tools and retouched artefacts: End-scraper 1; Side-scraper 1; Transversal knife 1; Notch 4; Denticulate 2; Varia 1. Total 10.

All pottery finds were from the IA. Flint artefacts were crude and large, from brown to violet flint, with some light-coloured patina similar to the other Yafit sites. No items indicating other periods were found; thus the assemblage is attributed to the IA.

Period: IA (general) – reasonable identification.

Site 110. 'Monument' (1)

Israel Grid 1937/1624, Elevation 240 m b.s.l.

The flint finds from this site were not found when the assemblages were processed.

Summary of Landscape Unit 29

In Landscape Unit 29, at 24 sites, 728 flint artefacts were collected. The area is semi-arid, and the elevation of the sites (except one) ranged from 170 m b.s.l. to 300 m b.s.l. Special attention should be paid to the increasing number of sites from the IA.

Review of the periods

Two Upper Palaeolithic sites from this period were identified as highly reasonable – Site 86 – Rujm es-Si'a and Site 104 – Yafit (4), both at elevations of 200–220 m b.s.l. At this elevation the area was flooded for a certain period by Lake Lisan (Bar Tov et al. 2002: fig. 7) and covered by lake deposits. The remnants of the period could be found on the present day surface only after being exposed by natural forces or by human activities. Sites from this period have already been found in nearby Wadi Ahmar (Goring-Morris 1980a).

Four Chalcolithic sites were identified at all levels. These sites were at elevations from 180 m b.s.l. to 300 m b.s.l., lower than the sites in Units 27 and 30. At the relevant period the elevation of the lake level was 370 m b.s.l. to 380 m b.s.l., and has continued to fall since then (Bar Tov et al. 2002: fig. 7). The area was never again flooded and covered by lake deposits; thus the artefacts remained exposed on the surface.

The tools at these sites were large and crude. The relatively high share of notches perhaps indicates some special activities. The absence of celt tools

Landscape Unit 29. Summary of sites by period. Identification according to flint artefacts			
Period	Highly reasonable	Reasonable	Possible
Middle Palaeolithic	-	-	-
Upper Palaeolithic	2	-	-
Epipalaeolithic	-	-	-
Neolithic	-	-	-
Chalcolithic	2	1	1
Early Bronze (EBA)	1	1	1
Intermediate Bronze (IBA)	1	-	-
Middle Bronze (MBA)	-	-	1
Late Bronze (LBA)	-	-	-
Bronze general (BA)	-	-	-
Iron I (IA I)	2	4	5
Iron II (IA II)	-	7	-
Iron general (IA)	1	2	2
Not defined	2		

Table A5: Landscape Unit 29 - Periods and identification level.

– the axe, the adze and sickle blades – is unusual for the period, but as no diagnostic flint artefacts from other periods were found and the pottery finds were from the Chalcolithic, the flint collections were also attributed to the Chalcolithic. In Landscape Unit 29 the Chalcolithic assemblages differ from those in Unit 27 located to the north. The axe, the adze and sickle blades were common tools in the agricultural communities of the Chalcolithic period (see reference in 'Location of sites' below), so their absence here needs an explanation. It makes sense that the economy of the communities here was based on herding.

Three sites were attributed to the EBA, but finds were few, and no diagnostic items were retrieved.

The IBA is represented at Site 81 – Khallet el-Fuleh. The few artefacts found were identified at the level of highly reasonable.

One site was attributed to the MBA at the level of possible identification.

All sites of the IA except one are located at elevations 180 m b.s.l. to 220 m b.s.l., relatively higher than the sites in Unit 28. Here too no diagnostic artefacts were found, but IA sherds dominated the pottery collections. The knapping techniques give evidence of a regular industry, but the artefacts were crude, and only a minimum of effort was invested in their production. The tools were identified as 'ad hoc' items (Rosen 1997: 34, 158-159). At these sites, located at low elevations similar to Landscape Unit 28, no sickle blades were found. It should be noted that the sickle blade is the most important tool in this period (see reference in 'Location of sites' below). At Site 93 (E.P. -189) from IA I, located 200 m b.s.l., a single geometric sickle blade without use-wear sheen was found. It was defined according to its shape, but no conclusions can be drawn from a single find. The missing sickle elements may indicate a population with a nomadic or semi-nomadic tradition whose existence is based on pasture. The continuity of this lifestyle during both phases of the IA is also evident in this landscape unit.

Landscape Unit 30

Site 111. Wadi el-Butmeh

Israel Grid 1913/1751, Elevation 70 m a.s.l.; 4 flint artefacts.

Artefacts without retouch: Flake 1; Blade 1.

Tools and retouched artefacts: Micro end-scraper 1; Retouched blade 1.

All artefacts were rather small, perhaps indicating a microlithic industry. The few finds do not allow a clear conclusion.

Period: Not defined.

Site 114. E.P. 103

Israel Grid 1914/1742, Elevation 95 m a.s.l.; 3 flint artefacts.

Artefacts without retouch: Flake 2.

Tools and retouched artefacts: Truncation 1.

Only three non-indicative artefacts were collected. 24% of the pottery were from IA II, and the flint was attributed to this period. The presence of flint artefacts confirms the continuity of flint utilization in this period.

Period: IA II – possible identification.

Site 115. Joret el-Kareireh

Israel Grid 1899/1734, Elevation 230 m a.s.l.; 206 flint artefacts (Fig. A15).

Artefacts without retouch: Core 4; Primary flake 13; Chunk 33; Chip 68; Flake 41; Blade 9; Bladelet 8; Microburin 2. Total 178.

Tools and retouched artefacts: End-scraper 2; Transversal scraper 1; Thumbnail scraper 2; Scraper-Burin 1; Side-scraper 3; Burin 2; Awl 1; Notch 4; Denticulate 2; Truncation 7; Retouched bladelet 1; Varia 2. Total 28.

The artefacts in this assemblage are very small, and only a few are 5 cm or longer. The cores are also small, and one of them is a bladelet core. Most blades and bladelets are broken, and some are only short fragments. Heat fractures and signs of burning are visible on many pieces. The four truncations were on short blade fragments. The collection did not include any diagnostic items, but the size of the artefacts, the presence of bladelets, the microburins and the relative large number of scrapers generally point to the Epipalaeolithic. At the site were many rock-hewn installations resembling Site 123 below, just a short distance away.

Period: Epipalaeolithic – reasonable identification.

Site 117. El-Medakakin

Israel Grid 1919/1738, Elevation 135 m a.s.l.

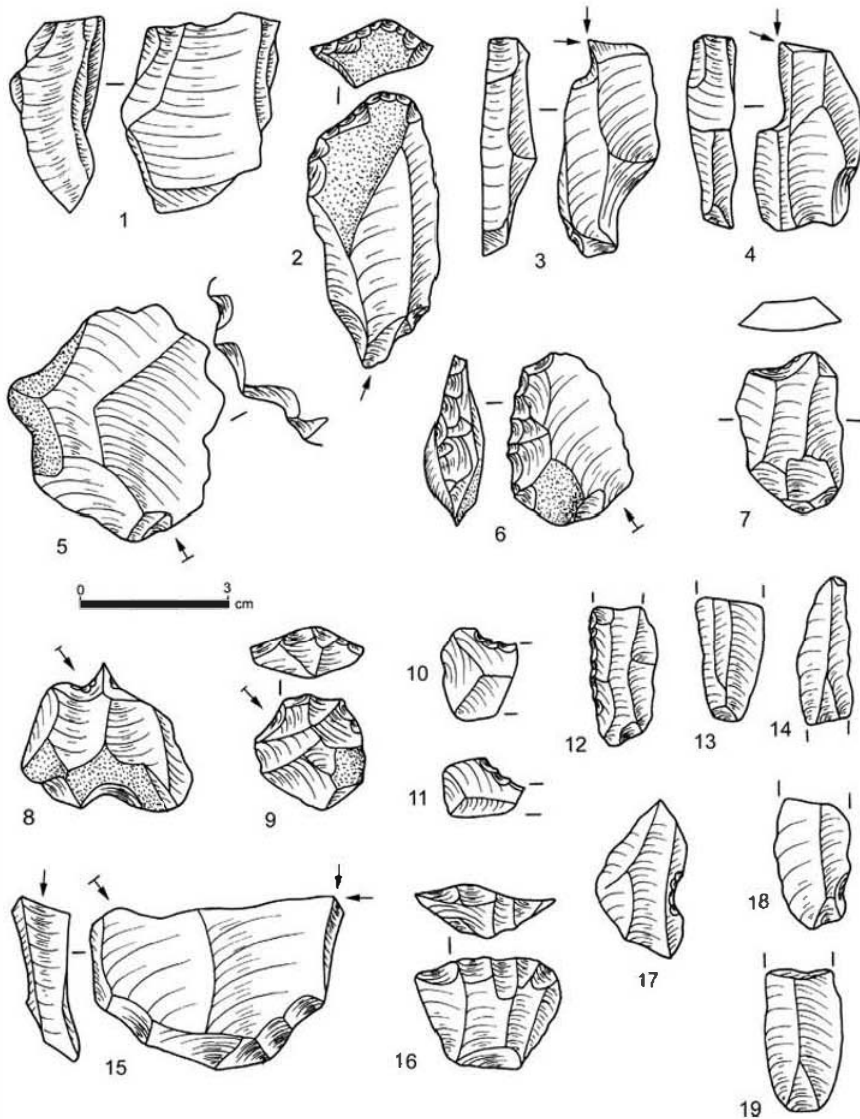
A small Mousterian flint scatter, 200 m south-west of Site 117, reported by Hovers and Bar-Yosef (1987: 79).

Period: Middle Palaeolithic – highly reasonable identification.

Site 123. Umm Sawaneh (5)

Israel Grid 1908/1720, Elevation 110 m a.s.l.; 11 flint artefacts (Fig. A15).

Artefacts without retouch: Core 2; Primary flake 7; Chunk 16; Chip 52; Flake 22; Blade 2; Bladelet 4; Microburin 1. Total 106.



A15. Joret el-Kareireh: 1-Bladelet core; 2-Scraper-burin; 3, 4-Burins; 5-Denticulate; 6-Side-scraper; 7-Notch; 8-Awl; 9-Thumbnail scraper; 10, 11-Microburins; 12-Retouched bladelet; 13, 14-Bladelets. Umm Sawaneh (5): 15-Burin; 16-Transversal scraper; 17-Notch; 18, 19-Bladelets.

Tools and retouched artefacts: End-scraper 1; Burin 2; Notch 2. Total 5.

Many artefacts were collected, but diagnostic items were few. Prominent in the assemblage was waste, mainly chips; there were few tools, and those only with marginal retouch. All blades and bladelets were broken. Heat fractures and signs of burning were visible on many pieces. Beyond doubt, most items are industrial products, but apparently the completed tools were removed. The microburin, bladelets, and the large amount of small waste indicate a micro-lithic industry. At the site were many rock-hewn installations resembling Site 115 above, which is located nearby.

Period: Epipalaeolithic – reasonable identification.

Site 126. Umm Sawaneh (3)

Israel Grid 1912/1722, Elevation 100 m a.s.l.; 162 flint artefacts (Fig. A17).

Artefacts without retouch: Core 3; Primary flake 5; Core trimming element 1; Chunk 56; Chip 58; Flake 16; Levallois flake 1; Levallois point 1; Blade 8. Total 149.

Tools and retouched artefacts: End-scraper 3; Side-scraper 2; Burin 2; Notch 4; Denticulate 1; Arrow head 1. Total 13.

The relatively large collection includes much waste material, evidence of local production. Most items were small, but some were extremely large. These large artefacts were made from good quality grey flint with a thick beige patina. Most small artefacts were bare of any patina, but there were some patinated pieces with secondary retouch without patina on the retouched areas.

The Levallois pieces and the scraper on a flake with a faceted base help to identify the period. These techniques and the side-scrapers are common in the Middle Palaeolithic. Many sites from this period were found in a nearby area – Ras Kharubeh (Winter 1996: 670-676). Furthermore, the pottery finds should not be disregarded – 48% from IA I and 31% from IA II indicate that some of the artefacts may originate from these periods. This assumption is supported by the fact that many items, mainly waste, were without patina.

Periods: Middle Palaeolithic – highly reasonable identification (part of the finds).

IA I – reasonable identification (part of the finds).

IA II – reasonable identification (part of the finds).

Site 127. Wadi e-Zeith (1)*Israel Grid 1910/1720, Elevation 100 m a.s.l.; 5 flint artefacts.*

Artefacts without retouch: Primary flake 1; Flake 2. Total 3.

Tools and retouched artefacts: Burin 1; Notch 1. Total 2.

The small collection does not include diagnostic items. All the pottery was from late historical periods, and does not provide any indication for the identification of the flint.

Period: Not defined.

Site 128 (1) 'Iraq er-Resifeh Plateau³*Israel Grid 1933/1722, Elevation 105 m a.s.l. (Fig. A16)*

1091 flint artefacts were found in four different collecting areas:

1. Israel Grid: 19297/17223, 129 flint artefacts.

Artefacts without retouch: Flake 85; Blade 10; Core 3; Levallois core 1; Core trimming element 1. Total 108.

Tools and retouched artefacts: Retouched flake 18; Retouched blade 2; Scraper 9. Total 29.

2. Israel Grid: 19319/17201, 405 flint artefacts.

Artefacts without retouch: Flake 271; Blade 40; Core 13; Levallois core 2; Levallois point 1; Levallois blade 3; Levallois flake 6; Core trimming element 3. Total 339.

Tools and retouched artefacts: Retouched flake 39; Retouched blade 9; Scraper 8; Retouched Levallois flake 1; Retouched Levallois point 3; Natural backed blade 5; Blade with pressure retouch 1. Total 66.

3. Israel Grid: 19350/17213, 212 flint artefacts.

Artefacts without retouch: Flake 109; Blade 24; Bladelet 1; Core 3; Levallois core 2; Levallois point 2; Levallois blade 7; Levallois flake 3. Total 151.

Tools and retouched artefacts: Retouched flake 25; Retouched blade 12; Scraper 10; Retouched Levallois flake 5; Retouched Levallois point 3; Natural backed blade 6. Total 61.

4. Israel Grid: 19309/17230, 345 flint artefacts.

Artefacts without retouch: Flake 193; Blade 53; Core 8; Levallois core 4; Levallois point 1; Levallois flake 3. Total 263.

Tools and retouched artefacts: Retouched flake 34; Retouched blade 28; Scraper

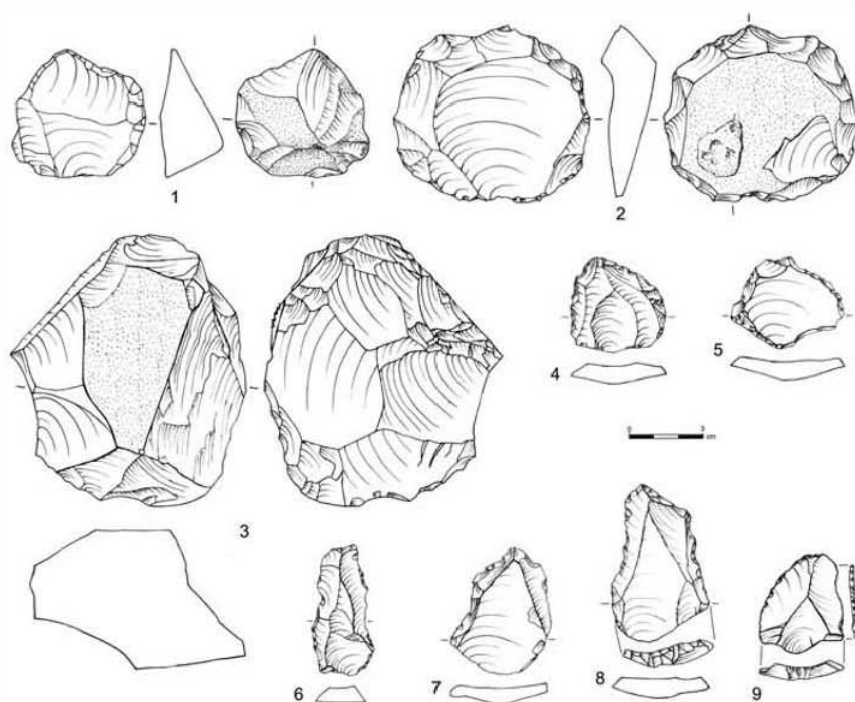
3 The flints from this site were analysed by Sonia Pinsky.

7; Retouched Levallois flake 2; Retouched Levallois point 1; Natural backed blade 10. Total 82.

These four collecting locations are part of a very large Middle Palaeolithic site or group of sites located on the 'Iraq er-Resifeh plateau. These locations were selected because of the apparent large amount of flint scatter in them. They represent only a selected portion of the site, and there are many flint scatters between them. On the other hand, there are areas between these locations with much lower densities of flint scatter, or with no flint at all. We therefore assume that the locations represent a long duration of presence on the plateau, and only a much more detailed prehistoric survey or test excavations will allow a more precise dating and determination of the area of this interesting site.

It is possible that the small Mousterian occurrence in Israel Grid 1942/1717, described by Hovers and Bar-Yosef (1987: 79), 1 km to the east of the centre of the site and on the eastern slope of the 'Iraq er-Resifeh plateau in the direction of Wadi Far'ah, is also part of this large site.

Period: Middle Palaeolithic – highly reasonable identification.



A16. 'Iraq er-Resifeh Plateau: 1-3-Levallois cores; 4, 5-Levallois flakes; 6-Levallois blade; 7-9-Levallois points.

Site 130. Jebel el-Mahjarah (2)

Israel Grid 1898/1716, Elevation 378 m a.s.l.; 11 flint artefacts.

Artefacts without retouch: Primary flake 2; Chunk 2; Flake 3; Blade 1. Total 8.

Tools and retouched artefacts: Retouched flake 1; Retouched blade 1; Notch 1. Total 3.

Diagnostic artefacts were not found. A few of the pottery items were from the MBA, LBA, and IA I; 65% were from IA II, and it is reasonable to attribute most of the flint artefacts to this period. No sickle blades were found at this site, located at a high elevation above the arid lowlands.

Period: IA II – reasonable identification (the bulk of the finds).

Site 132. Wadi ez-Zeith (2)

Israel Grid 1912/1719, Elevation 115 m a.s.l.; 13 flint artefacts.

Artefacts without retouch: Primary flake 1; Flake 6; Blade 2. Total 9.

Tools and retouched artefacts: Retouched blade 2; Denticulate 2. Total 4.

No artefacts indicating a specific period were found. Most of the pottery is from later irrelevant periods, but 40% were from IA II. It is reasonable to attribute the flint to this period.

Period: IA II – reasonable identification.

Site 137. Jebel el-Mahjarah (3)

Israel Grid 1910/1712, Elevation 280 m a.s.l.; 39 flint artefacts (Fig. A17)

Artefacts without retouch: Core 1; Primary flake 2; Chunk 2; Flake 13; Blade 10; Levallois point 3. Total 31.

Tools and retouched artefacts: Notch 1; Retouched blade 2; Retouched Levallois flake 1; Denticulate 2; Varia 2. Total 8.

Levallois artefacts with faceted bases are typical of the Middle Palaeolithic period. 27% of the pottery were from IA II, but no diagnostic artefacts from this period were found.

Periods: Middle Palaeolithic – highly reasonable identification (part of the finds).

IA II – possible identification (part of the finds).

Site 138. Jebel el-Mahjarah (1)

Israel Grid 1905/1710, Elevation 300 m a.s.l.; 27 flint artefacts (Fig. A18).

Artefacts without retouch: Primary flake 2; Chunk 1; Chip 1; Flake 4; Blade 10; Bladelet 1. Total 19.

Tools and retouched artefacts: End-scraper 1; Side-scraper 1; Backed knife 1; Notch 1; Retouched Levallois flake 2; Retouched Levallois point 2. Total 8.

Retouched Levallois flakes, retouched Levallois points, and the side-scraper,

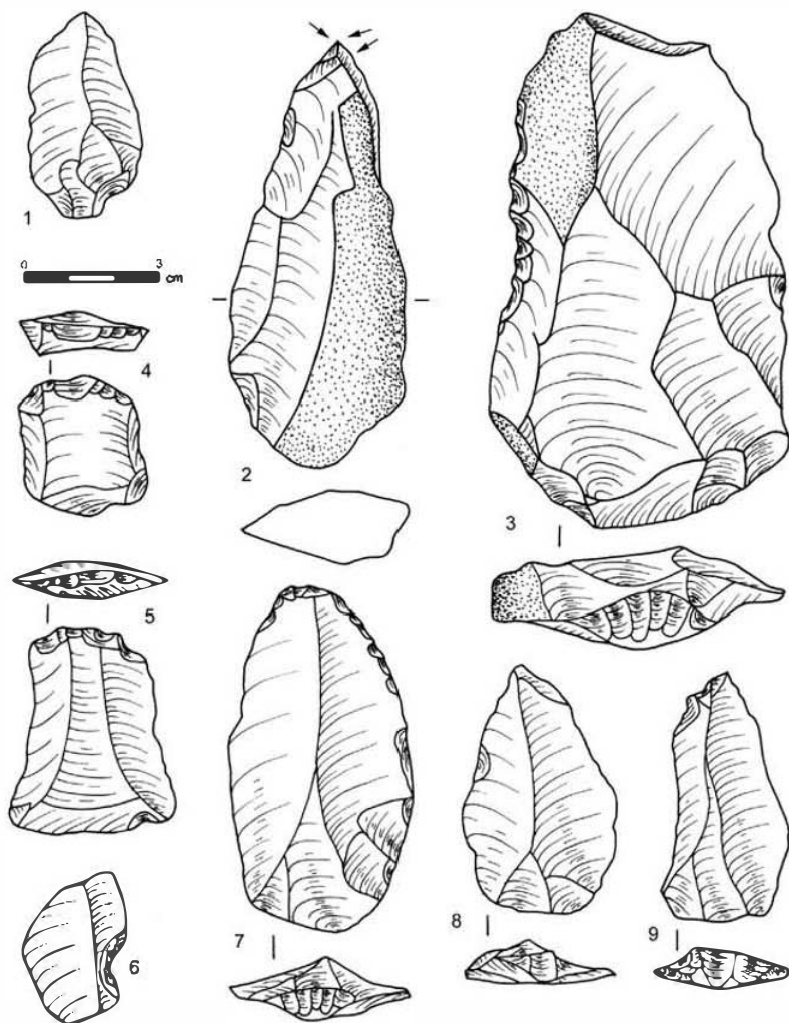


Fig. A17. Umm Sawaneh (3): 1-Levallois point; 2-Dihedral burin; 3-Side-scraper; 4, 5-End-scrappers; 6-Notch. Jebel el-Mahjarah (3): 7-Retouched Levallois flake; 8, 9-Levallois points.

all with faceted bases indicate the Middle Palaeolithic period. The other artefacts were not indicative. 18% of the pottery were from IA II, and some of the artefacts may be attributed to this period. The phenomenon that some Middle Palaeolithic sites were re-populated in the IA should be considered.

Periods: Middle Palaeolithic – highly reasonable identification (part of the finds).

IA II – possible identification (part of the finds).

Site 143. 'Ein Abu Daraj (1)

Israel Grid 1910/1700, Elevation 300 m a.s.l.; 46 flint artefacts (Fig. A18).

Artefacts without retouch: Core (Levallois) 2; Primary flake 9; Chunk 1; Chip 2; Flake 13; Blade 4; Bladelet 1. Total 32.

Tools and retouched artefacts: Thumbnail scraper 1; Side-scraper 1; Burin 1; Awl 1; Borer 1; Sickle blade 1; Notch 3; Denticulate 2; Retouched Levallois flake 2; Mousterian point 1. Total 14.

The Levallois artefacts provide firm evidence of the Middle Palaeolithic period. 50% of the pottery were from IA II, and it is most likely that a part of the flint finds are also from this period. The sickle blade confirms this assumption. This Middle Palaeolithic site was also re-populated in IA II.

Periods: Middle Palaeolithic – highly reasonable identification (part of the finds).

IA II – highly reasonable identification (part of the finds).

Site 146. Wadi Umm Hajar (2)

Israel Grid 1948/1705, Elevation 190 m b.s.l.; 13 flint artefacts.

Artefacts without retouch: Core trimming element 1; Chunk 3; Chip 1; Flake 4; Blade 1. Total 10.

Tools and retouched artefacts: End-scraper 1; Side-scraper 1; Retouched flake 1. Total 3.

No diagnostic items were found, and all pottery was from late historical periods.

Period: Not defined.

Site 149. E.P. 173 (1)

Israel Grid 1895/1692, Elevation 100 m a.s.l.; 88 flint artefacts.

Artefacts without retouch: Primary flake 13; Chunk 4; Chip 3; Flake 36; Blade 13; Bladelet 3; Ridge blade 1. Total 63.

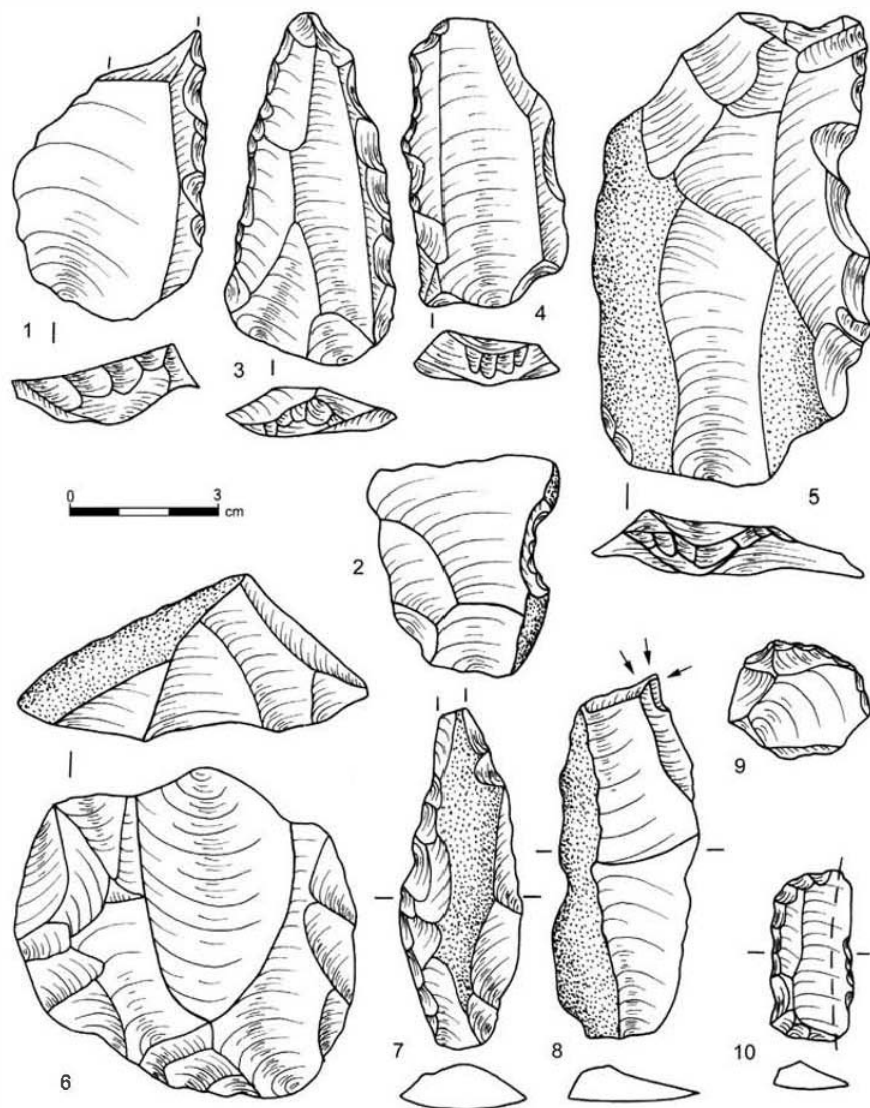


Fig. A18. Jebel el-Mahjarah (1) : 1-Side-scraper; 2-Notch. 'Ein Abu Daraj (1) : 3-Mousterian point; 4-Side-scraper; 5-Denticulate on Levallois flake; 6-Discoid Levallois core; 7-Borer; 8-Burin; 9-Thumbnail scraper; 10-Sickle blade.

Tools and retouched artefacts: End-scraper 3; Side-scraper 2; Burin 3; Retouched flake 7; Notch 6; Denticulate 2; Varia 2. Total 25.

The different types of flint found indicate a mixed assemblage from different periods. Seven artefacts with faceted bases and the side-scrappers are reasonable evidence of the Middle Palaeolithic period. Sites from this period are dispersed in this area. 6% of the pottery were from IA II, and a part of the collection may originate from this period.

Periods: Middle Palaeolithic – reasonable identification (part of the finds).

IA II – possible identification (part of the finds).

Sites 151. E.P. 195 (2) and 153. E.P. 195 (1)

Site 151: Israel Grid 1904/1694, Elevation 150 m a.s.l. (Fig. A19).

Site 153: Israel Grid 1902/1694, Elevation 195 m a.s.l. (Fig. A19).

378 flint artefacts (Fig. A18) were collected in the proximity of both sites, which are located close to each other on a hilltop and a lower ridge.

Artefacts without retouch: Core 2; Levallois core 5; Primary flake 45; Core trimming element 2; Chunk 40; Chip 26; Flake 100; Blade 26; Bladelet 1; Levallois flake 3; Levallois blade 2; Levallois point 2. Total 257.

Tools and retouched artefacts: End-scraper 11 (most were carinated); Rounded scraper 2; Side-scraper 7; Transversal scraper 3; Burin 5; Awl 10 (one on a Levallois flake); Borer 4; Retouched flake 16; Retouched blade 5 (one thick carinated blade); Retouched bladelet 1; Retouched Levallois flake 1; Retouched Levallois point 1; Naturally backed knife 4; Notch 28; Denticulate 5; Truncation 5; Varia 13. Total 121.

This relatively large collection attests to the density of the artefacts in this area. These were from good quality brown, tending to violet flint, and some had a thin light-coloured mottled patina. A similar thick patina could be distinguished on some broken Levallois pieces. Many of the artefacts indicate the period: Levallois cores, and Levallois items, including flakes, blades, points and side-scrappers, identify the Middle Palaeolithic. This is one of many sites of the period in this landscape unit and its vicinity (Ras Kharubeh, Winter 1996: 670–676). The continuity during the Upper Palaeolithic is indicated mainly by carinated end-scrappers and plain blades which are quite rare in the Middle Palaeolithic. The composition of the items attributed to the Upper Palaeolithic resembles that of 'Ein Aqev D31 (Marks 1976: 276) and Fazael X (Goring-Morris 1980a: 178). The reason for the scarcity of microliths lies apparently in the difficulty in finding small items on the ground without sieving. The data were insufficient to identify the stage within the period.

Periods: Middle Palaeolithic – highly reasonable identification (part of the finds).

Upper Palaeolithic – highly reasonable identification (part of the finds).

Site 159. Kh. Tawil (1)

Israel Grid 1868/1674, Elevation 330 m a.s.l.; 37 flint artefacts (Fig. A20).

Artefacts without retouch: Core 1; Primary flake 9; Chunk 6; Flake 6; Blade 2.

Total 24.

Tools and retouched artefacts: End-scraper 1; Awl (on broken sickle blade) 1;

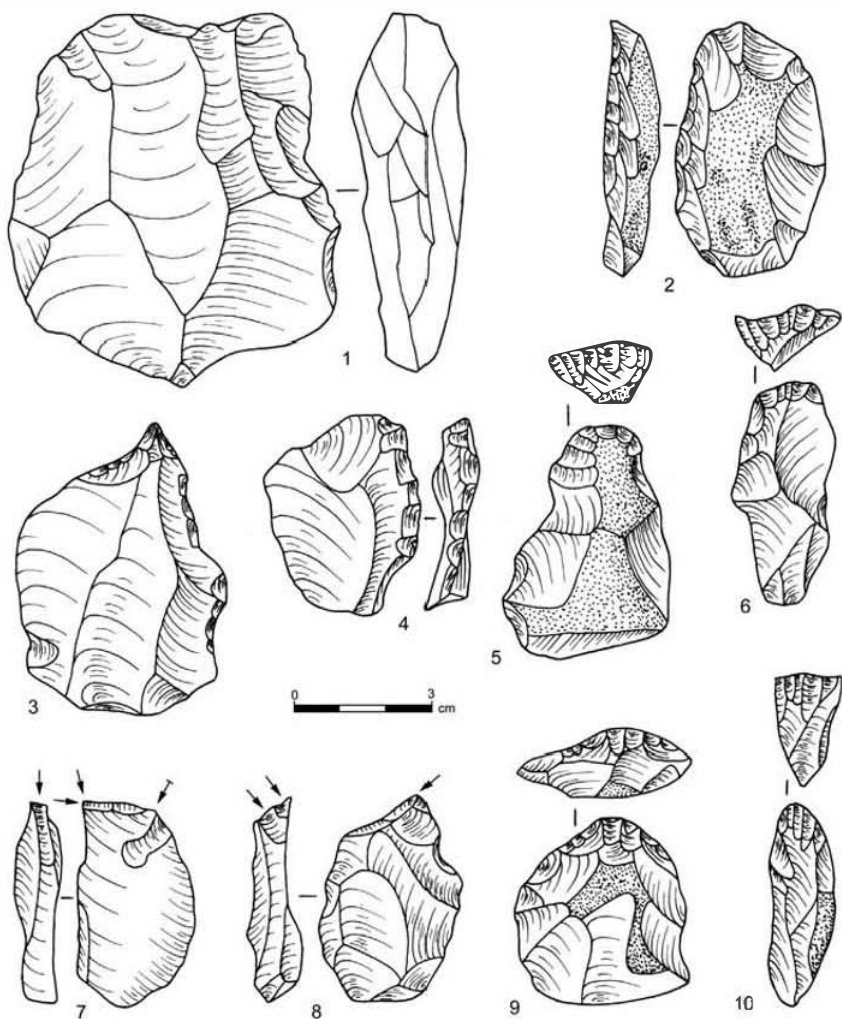


Fig. A19. E.P.195 (1) and (2): 1-Levallois core; 2, 4-Side-scrappers; 3-Retouched Levallois point; 5, 6, 9, 10-End-scrappers; 7, 8-Burins.

Retouched flake 1; Sickle blade 4; Notch 4; Varia 2. Total 13.

Only one item was patinated. The composition of the collection – core, waste, flakes, blades and retouched tools – attests to a full industrial process. 15% of the pottery were from MBA, 5% from IA I, 31% from IA II, and the rest from late historical periods. In contrast to sites at lower elevations, sickle blades with use-wear and sickle sheen were found, clear proof of agricultural activities. Two of these were prepared from blades, probably Canaanite blades attributed to EBA and IBA, and two geometric sickle elements, trapezoidal and rectangular, in use from MBA until the IA (Rosen 1997: 59-60 and fig. 3.19). The awl was also prepared from a broken sickle blade. The major part of the pottery from early periods is IA II; thus it makes sense to attribute most of the assemblage to this period. The geometric sickle elements support this thesis. The finds indicate a process of shifting from herders, with an economy based on herding in the lower and more arid areas, to farmers with an economy based on grain-growing at higher elevations. The verification of this thesis needs further research.

Periods: MBA – reasonable identification (part of the finds).

IA I – possible identification (part of the finds).

IA II – highly reasonable identification (part of the finds).

Site 160. Kh. Tawil (2) - el-Miniyah

Israel Grid 1870/1672, Elevation 340 m a.s.l.

Three retouched blades.

All three blades were broken, but the bases and bulbs of percussion prove intentional flaking. 87% of the pottery were from MBA, and it is reasonable to attribute the flint items similarly.

Period: MBA – reasonable identification.

Site 162. Merah es-Sahlat

Israel Grid 1895/1674, Elevation 90 m a.s.l.; 6 flint artefacts (Fig. A20).

Artefacts without retouch: Flake 1; Blade 1.

Tools and retouched artefacts: Transversal scraper 1; Side-scraper 1; Retouched blade 1; Denticulate 1. Total 4.

All artefacts were knapped. 16% of the pottery were from IA II, and the rest from later historical periods. In absence of indications of artefacts from periods earlier than IA II, the collection was attributed to this period.

Period: IA II – reasonable identification.